

**THE CONVEX NIVAT CONJECTURE:
A COMPLEXITY LOWER BOUND FOR STAR CONFIGURATIONS,
AND A REDUCTION FROM LOW CONVEX COMPLEXITY
TO STAR CONFIGURATIONS**

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ABSTRACT. Nivat’s conjecture asserts that a configuration $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ over a finite alphabet with $P_\xi(m, n) \leq mn$ for some $m, n \geq 1$ is periodic. Its convex form replaces the rectangle by an arbitrary non-empty finite lattice-convex window S , that is, one with $S = \text{Conv}(S) \cap \mathbb{Z}^2$. This paper contains two results.

The first occupies §§0–7: we call $\theta = \sum_{i=1}^m F_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ a *star configuration* when each F_i has a non-zero period $k_i v_i$, is not doubly periodic, and agrees with doubly periodic fields on the two half-planes $\{\pi_i < \ell_i\}$ and $\{\pi_i > r_i\}$, the primitive directions v_i being pairwise non-parallel; and we prove that $P_\theta(S) \geq |S| + 1$ for every non-empty finite lattice-convex S . The proof attaches to θ an *exceptional spectrum* Λ_i of roots of unity in each direction, a Laurent polynomial $A = \prod_i A_i(T^{v_i})$ that divides every annihilator of θ modulo constants, and the zonotope $Z = \text{Newt}(A) = \sum_i [0, d_i v_i]$. The zonotope then appears twice, with opposite signs: it is the exact cost of the affine relations carried by a window ($\dim U_S \geq |S| + 1 - |R_Z(S)|$), and it is the exact capacity for a family of quadratic characters that are independent modulo those relations. The two cancel, for every S and every Z .

The second result is a reduction (§8): the convex conjecture reduces to the first result, and therefore holds. Apart from the annihilator and periodic decomposition theorem of Kari and Szabados [11] and Colle’s results [2] on one-sided nonexpansive directions and on full periodicity over a region, every step of the chain is proved here in full. Appendix D proves in full the $\mathbb{F}_p$ and convex form of Szabados’ theorem. A further appendix reports a numerical verification of the main theorem and of the intermediate assertions of §§2–5, including a counterexample to the natural over-generalisation of Lemma 3.1.

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INTRODUCTION

The problem. Let $\mathcal{A}$ be a finite alphabet and $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ a configuration. For a finite $Q \subset \mathbb{Z}^2$ write $P_\xi(Q)$ for the number of distinct patterns $\xi|_{u+Q}$, $u \in \mathbb{Z}^2$, and $P_\xi(m, n) = P_\xi(\llbracket m \rrbracket \times \llbracket n \rrbracket)$ for the rectangular complexity, $\llbracket m \rrbracket = \{0, \dots, m-1\}$. The Morse–Hedlund theorem [13] says that a bi-infinite word is periodic as soon as its complexity function satisfies $P(n) \leq n$ for a single n . Nivat asked, in his 1997 ICALP address [14], whether the two-dimensional analogue holds:

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if $P_\xi(m, n) \leq mn$ for some $m, n \geq 1$, is ξ periodic?

The question is open. Partial results fall into two families. On the combinatorial and dynamical side, Sander–Tijdeman [16], Epifanio–Koskas–Mignosi [8] and Quas–Zamboni [15] proved periodicity under complexity bounds of the form $P_\xi(m, n) \leq mn/C$; Cyr and Kra [6] brought the constant down to 2 by way of nonexpansive subdynamics and settled in [7] the short-rectangle case $P_\xi(k, n) \leq kn$ for $k \leq 3$; Colle and Garibaldi [4] gave a version corrected by the size of the alphabet. On the algebraic side, Kari and Szabados [11] observed that low complexity forces the existence of a non-trivial annihilating Laurent polynomial and deduced that the configuration decomposes as a sum of periodic configurations over $\mathbb{Z}$; Szabados [17] settled the case of two components; Colle [2, 3] developed the interaction between minimal periodic decompositions and one-sided nonexpansive directions; and Colle and Garibaldi [5] recently proved, in the framework of $\mathbb{Z}_p$ -minimal periodic decompositions, a structure theorem — the orbit closure of an aperiodic configuration of low convex complexity contains an aperiodic configuration each of whose periodic components is fully periodic on two disjoint half-planes — and deduced from it the case $k \leq 4$. Kari and Moutot [9, 10] proved that the orbit closure of a configuration of low convex complexity contains a periodic configuration, and the related decidability statement for low-complexity tilings.

The *convex* form of the conjecture replaces the rectangle by an arbitrary window that is convex in the only sense available on a lattice: a finite $S \subset \mathbb{Z}^2$ is *lattice-convex* if $S = \text{Conv}(S) \cap \mathbb{Z}^2$. A configuration has *low convex pattern complexity* if $P_\xi(S) \leq |S|$ for some non-empty finite lattice-convex S . The convexity hypothesis is not decoration. Khetan [12] has constructed an exact cluster $F \subset \mathbb{Z}^2$ of cardinality 8 with full affine span and an F -tiling whose orbit closure contains no 1-periodic F -tiling; since every F -tiling has low complexity with respect to the window $-F$, this is a counterexample to the statement for non-convex windows of full affine span.

Results. Our results are two. The first is a complexity lower bound for star configurations, proved from scratch; the second reduces the convex Nivat conjecture to the first, quoting only results of [11] and [2].

Call $\theta: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ a *star configuration* if $\theta = \sum_{i=1}^m F_i$ with $m \geq 2$, where each F_i is periodic in a direction v_i , is not doubly periodic, and coincides with doubly periodic fields on two complementary half-planes orthogonal to nothing in particular but bounded by the lines $\pi_i = \det(v_i, \cdot) = \text{const}$, the primitive directions $v_1, \dots, v_m$ being pairwise non-parallel. The precise axioms (S1)–(S3) are in §0.1. Informally: m one-dimensionally periodic fields, each of which is doubly periodic outside a strip, superimposed in m pairwise distinct directions.

Theorem A (Theorem 7.3). *Let θ be a star configuration and let $S \subset \mathbb{Z}^2$ be non-empty, finite and lattice-convex. Then $P_\theta(S) \geq |S| + 1$.*

Theorem A is proved in §§0–7 using no input from the literature beyond the facts that $\mathbb{C}[T_1^\pm, T_2^\pm]$ is a unique factorisation domain, that Newton polygons are additive under multiplication (Ostrowski), the Chinese remainder theorem, Hilbert’s Nullstellensatz, and the existence of unimodular triangulations of lattice polygons.

Theorem B (Theorem 8.18). *Let $\mathcal{A}$ be finite and $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$. If $P_\xi(S) \leq |S|$ for some non-empty finite lattice-convex S , then ξ is periodic. Consequently, if $P_\xi(m, n) \leq mn$ for some $m, n \geq 1$, then ξ is periodic.*

Theorem B is a reduction of the convex conjecture to Theorem A (§8). The chain runs as follows: the annihilator and the periodic decomposition of Kari and Szabados [11]; the two

results of Colle [2] — a line that is one-sidedly nonexpansive in both directions, and an aperiodic configuration ϑ in the orbit closure which is fully periodic on a region; and then the two steps of this paper. The first is that each periodic component is fully periodic on one half-plane U_i (Proposition 8.9: extend the full periodicity over the region to a doubly periodic field, isolate the component with a difference operator Q_i , and run a finite-state argument). The second is that each component is fully periodic on a half-plane V_i on the far side (Theorem 8.12: push to the limit along the direction of some component; the limit is a sum of two periodic configurations in distinct directions, hence periodic by Theorem D.1 of Appendix D, and therefore every limit on the far side of the component under examination is doubly periodic; that there are only finitely many such limits then forces the far side itself to agree with a doubly periodic field). Finally Lemma 8.17 normalises such a ϑ into a star configuration, contradicting Theorem A. Theorem 8.1 — each component is fully periodic on two disjoint half-planes — coincides formally with the structure theorem of [5]; the proof given here is different from the one there and uses no minimality of the decomposition.

The shape of the proof of Theorem A. Let H_i be a period of F_i along v_i that also preserves every tail field, and let $D = \prod_{i=1}^m (T^{H_i} - 1)$.

A dichotomy (§1). Applying D to a function that depends on θ through a finite window kills everything except a bounded region, because outside the union of the pairwise intersections of m strips in pairwise distinct directions at most one component is still varying. If $D\mathcal{K}[\theta = a] \neq 0$ for some colour a , the theorem follows for *arbitrary* finite S , convex or not, by a one-line integral-domain argument. The rest of the paper is the remaining case $D\mathcal{K}[\theta = a] \equiv 0$.

The exceptional spectrum and its zonotope (§2). Pushing θ to infinity along $\pm H_i$ isolates the i -th component against a doubly periodic background, producing configurations $\Xi_i^\pm = F_i + B_i^\pm$ in the orbit closure together with their own pure tails $G_i^{\sigma,L}, G_i^{\sigma,R}$. The differences between them are invariant under H_i , hence decompose under the κ_i -th roots of unity; the set of roots that actually occur is the exceptional spectrum Λ_i , non-empty exactly because F_i is not doubly periodic. Set $A = \prod_i A_i(T^{v_i})$ with $A_i(t) = \prod_{\lambda \in \Lambda_i} (t - \lambda)$, and $Z = \text{Newt}(A) = \sum_i [0, d_i v_i]$, $d_i = |\Lambda_i|$. The key divisibility is Lemma 2.4: if $f(T)\eta$ is constant for a scalar encoding η of θ , then $A \mid f$. Counting, this says that a window S carries at most $|R_Z(S)|$ independent affine relations, where $R_Z(S) = \{r : r + Z \subseteq \text{Conv}(S)\}$: the *cost* of the zonotope.

The quadratic witness (§§3–6). Linear characters alone cannot close the gap, and they should not: the deficit $|R_Z(S)|$ is real. We produce $|R_Z(S)|$ further characters that are quadratic. For that one needs a single vector d with $D_z[\eta(z)\eta(z+d)] \neq 0$ and $d \in (Z - Z) \cap \mathbb{Z}^2$. Existence of some such d is §4; confinement to $Z - Z$ comes from a pair of recursions in the two variables (d, z) (§5) which say that the Laurent transform of the witness kills the ideal $(A(Y), A(X/Y))$ in $K[Y^\pm]$, $K = \mathbb{C}(X_1, X_2)$, whose quotient is spanned by monomials with exponents in $Z - Z$; and the passage from $d \in Z - Z$ to a *pair* of lattice points of Z differing by d is a unimodular triangulation argument (§6).

The cancellation (§7). Each $r \in R_Z(S)$ now yields a quadratic character Φ_r on the pattern set, and these are independent modulo the linear ones. The zonotope's capacity is exactly its cost, for every window and every zonotope; that is Remark 7.4.

Organisation. §0 fixes notation, the axioms of a star configuration, and the main theorem. §1 proves the dichotomy. §2 constructs Λ_i , A , Z and the dimension budget. §3 shows that A applied to the colour field of θ is a doubly periodic background. §4 produces a non-vanishing two-point difference, §5 confines it to $Z - Z$, and §6 converts that into a pair of lattice points of Z . §7 assembles Theorem A, and §8 deduces Theorem B: §8.1 sets up half-planes, regions and full periodicity; §8.2 collects the results that are quoted; §8.3 proves the first half-plane

(Proposition 8.9); §8.4 proves the second (Theorem 8.12); §8.5 normalises; §8.6 proves Theorem B. Appendix A is a one-page skeleton of the argument. Appendix B briefly describes, without proof and for the record only, a different route that was explored and is not used here. Appendix C reports the numerical verification. Appendix D proves the $\mathbb{F}_p$ and convex form of Szabados' theorem.

Dependencies. §§0–7 are self-contained. §8 quotes [11] (annihilators and periodic decompositions) and [2, Theorem 1.9 and Lemma 4.6]. Appendix D uses nothing beyond the one-dimensional Morse–Hedlund theorem [13].

0. NOTATION, HYPOTHESES AND THE MAIN THEOREM

0.1. Star configurations. Let p be a prime and $m \geq 2$. Let $v_1, \dots, v_m \in \mathbb{Z}^2$ be *primitive* (coprime coordinates) and *pairwise non-parallel*. Write

$$\pi_i(z) = \det(v_i, z) \quad (z \in \mathbb{Z}^2),$$

a surjective linear form $\mathbb{Z}^2 \rightarrow \mathbb{Z}$, surjective because v_i is primitive.

Call $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ *doubly periodic* if its period group $\text{Per}(G) = \{u \in \mathbb{Z}^2 : T^u G = G\}$ contains two linearly independent vectors; equivalently, $\text{Per}(G)$ is a finite-index sublattice of $\mathbb{Z}^2$.

Let $F_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ satisfy

- (S1) F_i has period $k_i v_i$ for some integer $k_i \geq 1$;
- (S2) F_i is *not* doubly periodic;
- (S3) there are doubly periodic fields L_i, R_i and integers $\ell_i \leq r_i + 1$ with

$$F_i = L_i \text{ on } \{\pi_i < \ell_i\}, \quad F_i = R_i \text{ on } \{\pi_i > r_i\}.$$

(If one starts with $\ell_i > r_i + 1$, shrinking ℓ_i to $r_i + 1$ keeps (S3).) We call $\{\pi_i < \ell_i\}$ and $\{\pi_i > r_i\}$ the *left* and *right tail regions* of component i , call L_i, R_i its *left* and *right tails*, and call $\{\ell_i \leq \pi_i \leq r_i\}$ the *transition strip* (possibly empty). We do not require $L_i = R_i$, do not require either to be constant, and do not require $k_i = 1$. Put

$$\theta = \sum_{i=1}^m F_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p,$$

and call θ a *star configuration*.

0.2. Complexity and the main theorem. For finite $Q \subset \mathbb{Z}^2$ let $\theta|_{u+Q}$ denote the function $q \mapsto \theta(u+q)$ and

$$P_\theta(Q) = \#\{\theta|_{u+Q} : u \in \mathbb{Z}^2\}.$$

Call a finite $S \subset \mathbb{Z}^2$ *lattice-convex* if $S = \text{Conv}(S) \cap \mathbb{Z}^2$.

Theorem T. *Let θ be a star configuration and let $S \subset \mathbb{Z}^2$ be non-empty, finite and lattice-convex. Then*

$$P_\theta(S) \geq |S| + 1.$$

Remark 0.1 (Periodic backgrounds). If a doubly periodic field B is added to θ , absorb B into F_1 and replace the tangential period of F_1 by a multiple that also preserves B ; such a multiple exists because $\text{Per}(B)$ has finite index, so $\text{Per}(B) \cap \mathbb{Z}k_1 v_1 \neq 0$. Then $F_1 + B$ is still not doubly periodic (otherwise $F_1 = (F_1 + B) - B$ would be), and $L_1 + B, R_1 + B$ are still doubly periodic. Hence Theorem T covers an arbitrary doubly periodic background.

0.3. Translations, differences and basic conventions. Write $T^u f(z) = f(z + u)$. A *Laurent polynomial* is a finite sum $f = \sum_u c_u T^u$ with complex coefficients unless stated otherwise, with support $\text{supp}(f) = \{u : c_u \neq 0\}$ and Newton polygon $\text{Newt}(f) = \text{Conv}(\text{supp}(f))$. The ring $\mathbb{C}[T_1^\pm, T_2^\pm]$ is an integral domain and a unique factorisation domain, and its units are exactly the monomials cT^u with $c \in \mathbb{C}^\times$.

Colour indicators, the encoding, the differences and all linear algebra take place in characteristic zero; only the summation of the original components happens in $\mathbb{F}_p$. In particular $\mathbb{K}[\theta = a]$ takes values in $\{0, 1\} \subset \mathbb{Z}$.

The common period lattice and the vectors H_i . Put

$$\Gamma := \bigcap_{j=1}^m (\text{Per}(L_j) \cap \text{Per}(R_j)),$$

a finite intersection of finite-index sublattices, hence of finite index; choose $N \in \mathbb{Z}_{>0}$ with $N\mathbb{Z}^2 \subseteq \Gamma$. For each i we have $Nk_i v_i \in \Gamma$, so we may put

$$H_i := \kappa_i v_i, \quad \kappa_i := \min\{\kappa \in k_i \mathbb{Z}_{>0} : \kappa v_i \in \Gamma\}.$$

Thus H_i preserves F_i (by (S1)) and preserves every L_j, R_j , $j = 1, \dots, m$.

The difference operator and its formal exponents. Put

$$D = \prod_{i=1}^m (T^{H_i} - 1), \quad I_a = \mathbb{K}[\theta = a], \quad e_\theta = (I_a)_{a \in \mathbb{F}_p}.$$

Then $D \neq 0$: each factor is non-zero and $\mathbb{C}[T^\pm]$ is an integral domain. For $C \subseteq [m] := \{1, \dots, m\}$ write $H_C = \sum_{i \in C} H_i$, and call

$$\mathcal{E} = \{H_C : C \subseteq [m]\}, \quad \mathcal{E}_i = \{H_C : C \subseteq [m] \setminus \{i\}\}$$

the *formal exponent sets*. For an arbitrary function g one has the formal expansion

$$(Dg)(z) = \sum_{C \subseteq [m]} (-1)^{m-|C|} g(z + H_C), \quad (0.1)$$

valid as an operator identity irrespective of whether the vectors H_C for different C coincide. Thus $\text{supp}(D) \subseteq \mathcal{E}$, and for $m \geq 3$ the inclusion may be strict, since distinct subsets may have equal sums (for instance if $H_3 = H_1 + H_2$). *Every pairing argument made on a formal expansion below is indexed by subsets C , never by the points H_C .* Likewise, for $Q_i := \prod_{j \neq i} (T^{H_j} - 1)$ one has $D = (T^{H_i} - 1)Q_i$ and

$$(Q_i g)(z) = \sum_{C \subseteq [m] \setminus \{i\}} (-1)^{m-1-|C|} g(z + H_C). \quad (0.2)$$

1. FINITE DIFFERENCES AND THE FIRST CASE

1.1. Differencing turns a local function into one of finite support.

Lemma 1.1. *Let $h(z) = \varphi(\theta|_{z+W})$ with $W \subset \mathbb{Z}^2$ finite. Then Dh has finite support.*

Proof. Put $M = \mathcal{E} + W$, a finite set, and $\text{diam}_i = \max_{s \in M} |\pi_i(s)|$. Define the *widened strips*

$$\Sigma_i = \{z : \ell_i - \text{diam}_i \leq \pi_i(z) \leq r_i + \text{diam}_i\},$$

and call direction i *active at the anchor z* if $z \in \Sigma_i$. If i is not active at z , then for every $s \in M$ the values $\pi_i(z + s)$ and $\pi_i(z)$ lie both in $\{\pi_i < \ell_i\}$ or both in $\{\pi_i > r_i\}$, so on $z + M$ the field F_i is identically one pure tail $\varepsilon_i \in \{L_i, R_i\}$.

Claim: if at most one direction is active at z , then $Dh(z) = 0$. Let J be the set of active directions, $|J| \leq 1$. Take $j_0 \in J$ if $J \neq \emptyset$, and $j_0 = 1$ otherwise. On $z + M$,

$$\theta = G + B, \quad G = \begin{cases} F_{j_0}, & J = \{j_0\} \\ 0, & J = \emptyset \end{cases}, \quad B = \sum_{j \notin J} \varepsilon_j,$$

where B is doubly periodic and preserved by every H_l , and G is preserved by H_{j_0} by (S1). Hence $G + B$ is preserved by H_{j_0} . Expand $D = (T^{H_{j_0}} - 1)Q_{j_0}$ using (0.2):

$$Dh(z) = \sum_{C \subseteq [m] \setminus \{j_0\}} (-1)^{m-1-|C|} [h(z + H_C + H_{j_0}) - h(z + H_C)].$$

Fix C . Since H_C and $H_C + H_{j_0}$ both lie in $\mathcal{E}$, the two windows $z + H_C + W$ and $z + H_C + H_{j_0} + W$ are contained in $z + M$, where $\theta = G + B$. So for $y \in W$,

$\theta(z + H_C + H_{j_0} + y) = (G + B)(z + H_C + y + H_{j_0}) = (G + B)(z + H_C + y) = \theta(z + H_C + y)$, so the two W -patterns agree and the bracket vanishes. This proves the claim.

Conclusion. Therefore $\text{supp}(Dh) \subseteq \bigcup_{i \neq j} (\Sigma_i \cap \Sigma_j)$. As $v_i \nparallel v_j$, the intersection of two strips of bounded width in non-parallel directions is bounded, so Dh has finite support. $\square$

1.2. The integral-domain argument.

Lemma 1.2. *Let $J: \mathbb{Z}^2 \rightarrow \mathbb{C}$ be non-zero with finite support and let f be a non-zero Laurent polynomial. Then $f(T)J \neq 0$.*

Proof. Write $\hat{J}(X) = \sum_z J(z)X^{-z} \in \mathbb{C}[X_1^\pm, X_2^\pm]$, which is non-zero. For $f = \sum_u c_u T^u$,

$$\sum_z (f(T)J)(z)X^{-z} = \sum_u c_u \sum_z J(z + u)X^{-z} = \sum_u c_u \sum_{z'} J(z')X^{-z'+u} = \left(\sum_u c_u X^u \right) \hat{J}(X).$$

Both factors are non-zero Laurent polynomials and the ring is an integral domain, so the product is non-zero. $\square$

1.3. The first case.

Proposition 1.3. *If $DI_a \neq 0$ for some $a \in \mathbb{F}_p$, then $P_\theta(S) \geq |S| + 1$ for every non-empty finite $S \subset \mathbb{Z}^2$; no convexity is needed.*

Proof. Suppose $P_\theta(S) \leq |S|$. Write $\mathcal{L} = \{\theta|_{u+S} : u \in \mathbb{Z}^2\}$, so $|\mathcal{L}| \leq |S|$. On $\mathcal{L}$ consider the $|S| + 1$ functions

$$\mathbf{1}; \quad \gamma \mapsto \mathbb{K}[\gamma(s) = a] \quad (s \in S).$$

They lie in a space of dimension $\leq |\mathcal{L}| \leq |S|$, hence are linearly dependent: there are rationals $c_0, (c_s)_{s \in S}$, not all zero, with

$$\sum_{s \in S} c_s \mathbb{K}[\gamma(s) = a] = c_0 \quad \text{for all } \gamma \in \mathcal{L}.$$

Substituting $\gamma = \theta|_{u+S}$ gives $q(T)I_a = c_0$ with $q = \sum_s c_s T^s$. If $q = 0$ then all $c_s = 0$ and hence $c_0 = 0$, a contradiction; so $q \neq 0$.

Apply D to both sides. The right-hand side gives $Dc_0 = 0$, since a constant is killed by $T^{H_1} - 1$; the left-hand side gives $q(T)DI_a$. By Lemma 1.1 and the hypothesis, DI_a is non-zero with finite support, so $q(T)DI_a \neq 0$ by Lemma 1.2. Contradiction. $\square$

From now on we assume

$$DI_a = 0 \quad \text{for every } a \in \mathbb{F}_p. \tag{1.1}$$

Consequently $Dw(\theta) = \sum_a w(a)DI_a = 0$ for every function $w: \mathbb{F}_p \rightarrow \mathbb{Z}$.

Remark 1.4 (The dichotomy does not depend on the choice of the H_i). Replacing H_i by a positive multiple $n_i H_i$ produces $D' = \prod_i (T^{n_i H_i} - 1)$, which is a multiple of D because $T^{nH} - 1 = (T^H - 1)(1 + T^H + \dots + T^{(n-1)H})$. Hence $DI_a = 0$ implies $D'I_a = 0$; conversely, if $DI_a \neq 0$ then it is non-zero of finite support by Lemma 1.1, and Lemma 1.2 gives $D'I_a = (D'/D)(T)DI_a \neq 0$. So whether (1.1) holds does not depend on the particular κ_i chosen, as long as H_i preserves all the tail fields.

2. EXCEPTIONAL SPECTRA, THE ENCODING, AND THE DIMENSION BUDGET FOR AFFINE RELATIONS

2.1. Isolating configurations. Fix i . For $j \neq i$ we have $\pi_j(H_i) = \det(v_j, H_i) = \kappa_i \det(v_j, v_i) \neq 0$, so $\pi_j(nH_i) \rightarrow \pm\infty$; hence on any fixed finite set F_j is eventually identically equal, as $n \rightarrow +\infty$, to

$$\varepsilon_j^+ = \begin{cases} R_j, & \pi_j(H_i) > 0, \\ L_j, & \pi_j(H_i) < 0, \end{cases}$$

while F_i is preserved by H_i . So $T^{nH_i}\theta$ stabilises on every finite set, with limit

$$\Xi_i^+ = F_i + B_i^+, \quad B_i^+ = \sum_{j \neq i} \varepsilon_j^+.$$

Letting $n \rightarrow -\infty$ gives $\Xi_i^- = F_i + B_i^-$, with the opposite tail choices: ε_j^- is L_j when $\pi_j(H_i) > 0$ and R_j otherwise. Both belong to the orbit closure $X_\theta = \overline{\{T^u \theta : u \in \mathbb{Z}^2\}}$ in the product topology. Since $\kappa_i > 0$ we have $\text{sign } \pi_j(H_i) = \text{sign } \pi_j(v_i)$, so that

$$\varepsilon_j^\sigma = \begin{cases} R_j, & \sigma \pi_j(v_i) > 0, \\ L_j, & \sigma \pi_j(v_i) < 0. \end{cases} \quad (2.0)$$

By (S3),

$$\Xi_i^\sigma = G_i^{\sigma,L} := L_i + B_i^\sigma \text{ on } \{\pi_i < \ell_i\}, \quad \Xi_i^\sigma = G_i^{\sigma,R} := R_i + B_i^\sigma \text{ on } \{\pi_i > r_i\},$$

and $G_i^{\sigma,L}, G_i^{\sigma,R}$ are doubly periodic and preserved by Γ . They too belong to X_θ : take (v_i, u_i) a basis of $\mathbb{Z}^2$ with $\det(v_i, u_i) = 1$ and put $u = Nu_i \in \Gamma$, so that $\pi_i(u) = N > 0$. For fixed z and n large, $\pi_i(z + nu) > r_i$, whence

$$(T^{nu} \Xi_i^\sigma)(z) = R_i(z + nu) + B_i^\sigma(z + nu) = R_i(z) + B_i^\sigma(z) = G_i^{\sigma,R}(z),$$

that is, $T^{nu} \Xi_i^\sigma \rightarrow G_i^{\sigma,R}$; using $-u$ instead gives $G_i^{\sigma,L}$.

We record that H_i preserves Ξ_i^σ , and every H_j preserves each of $G_i^{\sigma,L}, G_i^{\sigma,R}$, the latter being sums of tails.

2.2. Spectral projections, exceptional difference fields, and Λ_i . Take an integral basis (v_i, u_i) with $\det(v_i, u_i) = 1$ and write $z = sv_i + tu_i$; then $\pi_i(z) = t$ and $s = \det(z, u_i)$. Call $\{t = \text{const}\}$ a *transverse row*. Recall $H_i = \kappa_i v_i$.

Lemma 2.0 (Spectral projection). *Let $d: \mathbb{Z}^2 \rightarrow \mathbb{C}$ satisfy $T^{H_i} d = d$. For $\lambda^{\kappa_i} = 1$ put*

$$P_\lambda := \frac{1}{\kappa_i} \sum_{k=0}^{\kappa_i-1} \lambda^{-k} T^{kv_i}, \quad \widehat{d}_t(\lambda) := \frac{1}{\kappa_i} \sum_{k=0}^{\kappa_i-1} \lambda^{-k} d(kv_i + tu_i).$$

Then:

- (1) $(P_\lambda d)(sv_i + tu_i) = \lambda^s \widehat{d}_t(\lambda)$, and $\sum_\lambda P_\lambda d = d$;
- (2) $T^{v_i} P_\lambda d = \lambda P_\lambda d$;

- (3) P_λ is a Laurent polynomial in T , hence commutes with every T^u and with every Laurent polynomial $f(T)$;
 (4) for $f(T) = \sum_{\alpha, \beta} f_{\alpha\beta} T^{\alpha v_i + \beta u_i}$,

$$(f(T)P_\lambda d)(sv_i + tu_i) = \lambda^s \sum_{\alpha, \beta} f_{\alpha\beta} \lambda^\alpha \hat{d}_{t+\beta}(\lambda) = \lambda^s [f(\lambda, Y) \hat{d}_\bullet(\lambda)](t),$$

where $f(\lambda, Y) := \sum_{\alpha, \beta} f_{\alpha\beta} \lambda^\alpha Y^\beta$ acts, as a Laurent polynomial in the single variable Y , on functions of t , with Y the shift $t \mapsto t + 1$.

Proof. Parts 1 and 2 are the discrete Fourier decomposition on $\mathbb{Z}/\kappa_i$, carried out row by row: $(P_\lambda d)(sv_i + tu_i) = \frac{1}{\kappa_i} \sum_k \lambda^{-k} d((s+k)v_i + tu_i) = \lambda^s \cdot \frac{1}{\kappa_i} \sum_{k'} \lambda^{-k'} d(k'v_i + tu_i)$, re-indexing by κ_i -periodicity, while $\sum_\lambda \lambda^{s-k} = \kappa_i \mathbb{1}[k \equiv s]$ gives completeness. Part 3 is clear, and part 4 is a direct computation from part 1 and the definition of $T^{\alpha v_i + \beta u_i}$. $\square$

Say that λ occurs in d if $P_\lambda d \neq 0$, that is, if $\hat{d}_t(\lambda) \neq 0$ for some row t .

Define the *exceptional difference fields*

$$d_{i, \sigma, \epsilon, a}(z) = \mathbb{1}[\Xi_i^\sigma(z) = a] - \mathbb{1}[G_i^{\sigma, \epsilon}(z) = a] \in \{-1, 0, 1\} \quad (\sigma \in \{+, -\}, \epsilon \in \{L, R\}, a \in \mathbb{F}_p).$$

(P1) $d_{i, \sigma, R, a} \equiv 0$ on $\{\pi_i > r_i\}$, and $d_{i, \sigma, L, a} \equiv 0$ on $\{\pi_i < \ell_i\}$.

(P2) Both Ξ_i^σ and $G_i^{\sigma, \epsilon}$ are preserved by H_i , so $T^{H_i} d_{i, \sigma, \epsilon, a} = d_{i, \sigma, \epsilon, a}$ and Lemma 2.0 applies.

Put

$$\Lambda_i = \{\lambda : \lambda^{\kappa_i} = 1, \exists \sigma, \epsilon, a \text{ such that } \lambda \text{ occurs in } d_{i, \sigma, \epsilon, a}\}.$$

Lemma 2.1. $\Lambda_i \neq \emptyset$.

Proof. Otherwise every $d_{i, \sigma, \epsilon, a}$ vanishes identically, by part 1 of Lemma 2.0. Taking $\sigma = +$ and $\epsilon = R$ gives $\Xi_i^+ = G_i^{+, R}$; subtracting B_i^+ gives $F_i = R_i$, so F_i is doubly periodic, contradicting (S2). $\square$

Put

$$A_i(t) = \prod_{\lambda \in \Lambda_i} (t - \lambda), \quad d_i = \deg A_i = |\Lambda_i| \geq 1, \quad (2.1)$$

$$A(T) = \prod_{i=1}^m A_i(T^{v_i}), \quad Z = \sum_{i=1}^m [0, d_i v_i] \subset \mathbb{R}^2. \quad (2.2)$$

(P3) For all σ, ϵ, a one has $A_i(T^{v_i}) d_{i, \sigma, \epsilon, a} = 0$. Indeed, by Lemma 2.0 only the terms with $\lambda \in \Lambda_i$ in $d = \sum_\lambda P_\lambda d$ are non-zero, and $A_i(T^{v_i}) P_\lambda d = A_i(\lambda) P_\lambda d = 0$.

The elements of Λ_i are distinct κ_i -th roots of unity, so A_i is monic with $A_i(0) = \prod_\lambda (-\lambda) \neq 0$. Hence

$$\text{Newt}(A_i(T^{v_i})) = [0, d_i v_i], \quad \text{Newt}(A) = \sum_i [0, d_i v_i] = Z, \quad (2.3)$$

the second equality by Minkowski additivity of Newton polygons (Ostrowski: over an integral domain, $\text{Newt}(fg) = \text{Newt}(f) + \text{Newt}(g)$).

Remark 2.1' (Galois stability of Λ_i). Each $d_{i, \sigma, \epsilon, a}$ is integer-valued, so for every $\tau \in \text{Gal}(\mathbb{Q}(\zeta_{\kappa_i})/\mathbb{Q})$ one has $\tau(\hat{d}_t(\lambda)) = \hat{d}_t(\tau\lambda)$. Hence λ occurs if and only if $\tau\lambda$ occurs, Λ_i is a union of Galois orbits, A_i is a product of distinct cyclotomic polynomials, $A_i \in \mathbb{Z}[t]$ and $A \in \mathbb{Z}[T^\pm]$. Nothing below depends on this (Lemma 2.5 handles the coefficient field by invariance of rank under field extension), but it does mean that every object of §2 can be discussed over $\mathbb{Q}$.

2.3. A spectrum-preserving encoding.

Lemma 2.2. *There is an injection $w: \mathbb{F}_p \rightarrow \mathbb{Z}$ such that for every i and every $\lambda \in \Lambda_i$, the scalar difference field*

$$\delta_{i,\sigma,\epsilon} := w(\Xi_i^\sigma) - w(G_i^{\sigma,\epsilon}) = \sum_a w(a) d_{i,\sigma,\epsilon,a}$$

still has λ occurring in it, for some pair σ, ϵ .

Proof. Regard $(w(a))_{a \in \mathbb{F}_p}$ as a vector in $\mathbb{Q}^p$. For each $\lambda \in \Lambda_i$ there are, by definition, σ, ϵ, t for which the vector $(\widehat{d_{i,\sigma,\epsilon,a}})_t(\lambda) \in \mathbb{C}^p$, indexed by a , is non-zero. By linearity of the Fourier transform,

$$(\widehat{\delta_{i,\sigma,\epsilon}})_t(\lambda) = \sum_a w(a) (\widehat{d_{i,\sigma,\epsilon,a}})_t(\lambda)$$

is a non-zero complex linear form in w ; its zero set inside $\mathbb{Q}^p$ is a proper $\mathbb{Q}$ -subspace of $\mathbb{Q}^p$, since otherwise the form would vanish on a $\mathbb{Q}$ -basis of $\mathbb{Q}^p$ and hence identically. There are $\sum_i |\Lambda_i|$ such conditions. Injectivity excludes a further $\binom{p}{2}$ proper hyperplanes $\{w(a) = w(a')\}$. Finitely many proper subspaces cannot cover $\mathbb{Q}^p$, so a rational w satisfying all the conditions exists; multiplying by a common denominator makes it integer-valued without affecting non-vanishing or injectivity. $\square$

Fix such a w and put $\eta = w(\theta): \mathbb{Z}^2 \rightarrow \mathbb{Z}$. By (1.1), $D\eta = 0$.

2.4. One-sided vanishing.

Lemma 2.3. *Let $d: \mathbb{Z}^2 \rightarrow \mathbb{C}$ satisfy $T^{H_i}d = d$, vanish identically on a half-plane $\{\pi_i > c\}$ or on a half-plane $\{\pi_i < c\}$, and let λ occur in d . If a Laurent polynomial f satisfies $f(T)d = 0$, then $(T^{v_i} - \lambda) \mid f$.*

Proof. By part 3 of Lemma 2.0, $f(T)P_\lambda d = P_\lambda f(T)d = 0$. Write $c_\lambda(t) = \widehat{d}_t(\lambda)$, so $c_\lambda(t) = 0$ whenever d vanishes on row t , and $g(Y) := f(\lambda, Y) = \sum_\beta g_\beta Y^\beta$. By part 4 of Lemma 2.0,

$$0 = (f(T)P_\lambda d)(sv_i + tu_i) = \lambda^s \sum_\beta g_\beta c_\lambda(t + \beta) \quad \text{for all } s, t,$$

that is, $g(Y)c_\lambda = 0$; and $c_\lambda \not\equiv 0$ because λ occurs.

Case: d vanishes on $\{\pi_i > c\}$. Then $c_\lambda(t) = 0$ for $t > c$, so $t_0 := \max\{t : c_\lambda(t) \neq 0\}$ exists. If $g \neq 0$, let $\beta_- = \min\{\beta : g_\beta \neq 0\}$ and evaluate at $t = t_0 - \beta_-$: for $\beta > \beta_-$ one has $t_0 - \beta_- + \beta > t_0$ and c_λ vanishes there, so only the term $\beta = \beta_-$ survives, giving $g_{\beta_-} c_\lambda(t_0) = 0$, a contradiction.

Case: d vanishes on $\{\pi_i < c\}$. Symmetrically, take $t_0 = \min\{t : c_\lambda(t) \neq 0\}$ and $\beta_+ = \max\{\beta : g_\beta \neq 0\}$ and evaluate at $t = t_0 - \beta_+$; only the term $\beta = \beta_+$ survives.

So $g \equiv 0$. Write f as an element of $\mathbb{C}[X^\pm, Y^\pm]$ with $X = T^{v_i}$ and $Y = T^{u_i}$; since (v_i, u_i) is a basis of $\mathbb{Z}^2$ this is an automorphism of the Laurent ring. With $f = \sum_\beta Y^\beta f_\beta(X)$, the identity $g \equiv 0$ says $f_\beta(\lambda) = 0$ for every β , whence $(X - \lambda) \mid f_\beta(X)$ — a one-variable Laurent polynomial vanishing at a non-zero point is divisible by $X - \lambda$ — and therefore $(X - \lambda) \mid f$. $\square$

2.5. Divisibility.

Lemma 2.4. *Let f be a non-zero Laurent polynomial with complex coefficients such that $f(T)\eta = c$ is constant. Then $A \mid f$.*

Proof. (a) The relation passes to the orbit closure. The map $z \mapsto (f(T)\eta)(z) - c$ depends on θ only through its values on $z + \text{supp}(f)$, so it is a local function of θ . It vanishes on every translate of θ ; local functions are continuous for the product topology, so it vanishes on every member χ of X_θ : $f(T)w(\chi) = c$.

(b) Subtraction. Take $\chi = \Xi_i^\sigma$ and $\chi = G_i^{\sigma, \epsilon}$, both in X_θ by §2.1, and subtract the two identities: $f(T)\delta_{i, \sigma, \epsilon} = 0$.

(c) One prime factor at a time. Fix $\lambda \in \Lambda_i$. By Lemma 2.2 there are σ, ϵ with λ occurring in $\delta_{i, \sigma, \epsilon}$; by (P1) this difference field vanishes on a half-plane; by (P2) it is preserved by H_i . Lemma 2.3 gives $T^{v_i} - \lambda \mid f$.

(d) Assembling. Under the automorphism $X = T^{v_i}$, $Y = T^{u_i}$ the element $T^{v_i} - \lambda$ becomes $X - \lambda$, and $\mathbb{C}[X^\pm, Y^\pm]/(X - \lambda) \cong \mathbb{C}[Y^\pm]$ is an integral domain, so $T^{v_i} - \lambda$ is prime. Primes coming from different directions $i \neq k$ are non-associate: being associate would force $T^{v_i} - \lambda = cT^u(T^{v_k} - \mu)$, and comparing supports gives $\{0, v_i\} = \{u, u + v_k\}$, hence $v_i = \pm v_k$, contradicting non-parallelism. Primes from the same direction with $\lambda \neq \lambda'$ are also non-associate: comparing supports forces $u = 0$ and then comparing coefficients forces $\lambda = \lambda'$. In a unique factorisation domain, pairwise non-associate primes each dividing f have their product dividing f : $A = \prod_i \prod_{\lambda \in \Lambda_i} (T^{v_i} - \lambda) \mid f$. $\square$

2.6. The dimension budget for affine relations. For a lattice-convex S put

$$R_Z(S) = \{r \in \mathbb{Z}^2 : r + Z \subseteq \text{Conv}(S)\},$$

$$U_S = \text{span}_{\mathbb{Q}}\{\mathbf{1}, \gamma \mapsto w(\gamma(s)) \ (s \in S)\} \subseteq \{\mathcal{L} \rightarrow \mathbb{Q}\}.$$

Lemma 2.5. $\dim_{\mathbb{Q}} U_S \geq |S| - |R_Z(S)| + 1$.

Proof. U_S is generated by $|S| + 1$ elements, so $\dim_{\mathbb{Q}} U_S = (|S| + 1) - \dim_{\mathbb{Q}} \mathcal{R}$, where

$$\mathcal{R} = \left\{ (c_0, (c_s)_{s \in S}) \in \mathbb{Q}^{1+|S|} : \sum_{s \in S} c_s \eta(u + s) = c_0 \text{ for all } u \in \mathbb{Z}^2 \right\}.$$

$\mathcal{R}$ is the $\mathbb{Q}$ -solution space of a (possibly infinite) homogeneous linear system with integer coefficients; writing $\mathcal{R}_{\mathbb{C}}$ for its $\mathbb{C}$ -solution space, $\mathcal{R}_{\mathbb{C}} = \mathcal{R} \otimes_{\mathbb{Q}} \mathbb{C}$ and $\dim_{\mathbb{C}} \mathcal{R}_{\mathbb{C}} = \dim_{\mathbb{Q}} \mathcal{R}$, the dimension of the solution space of a rational linear system being unchanged by extension of the base field.

Let $(c_0, (c_s)) \in \mathcal{R}_{\mathbb{C}}$ be non-zero and put $f = \sum_s c_s T^s$. If $f = 0$ then $c_0 = 0$ and the element is zero; so $f \neq 0$, and $f(T)\eta = c_0$ with $\text{supp}(f) \subseteq S$. By Lemma 2.4, valid for complex f , we may write $f = Ag$ with $g \neq 0$. By (2.3) and Newton-polygon additivity,

$$\text{Newt}(f) = \text{Newt}(A) + \text{Newt}(g) = Z + \text{Newt}(g).$$

From $\text{supp}(f) \subseteq S$ we get $\text{Newt}(f) \subseteq \text{Conv}(S)$, so for every $r \in \text{supp}(g)$,

$$r + Z \subseteq \text{Newt}(g) + Z = \text{Newt}(f) \subseteq \text{Conv}(S),$$

that is, $r \in R_Z(S)$. Hence $\text{supp}(g) \subseteq R_Z(S)$.

The map $\mathcal{R}_{\mathbb{C}} \rightarrow \{g : \text{supp}(g) \subseteq R_Z(S)\}$, $(c_0, f) \mapsto g = f/A$, is $\mathbb{C}$ -linear and injective (c_0 is determined by f , $A \neq 0$, and division in an integral domain is unique), so

$$\dim_{\mathbb{Q}} \mathcal{R} = \dim_{\mathbb{C}} \mathcal{R}_{\mathbb{C}} \leq |R_Z(S)|. \quad \square$$

Remark 2.6. By lattice-convexity, $r \in R_Z(S)$ implies $(r + Z) \cap \mathbb{Z}^2 \subseteq \text{Conv}(S) \cap \mathbb{Z}^2 = S$. If $R_Z(S) = \emptyset$, then Lemma 2.5 gives $\dim U_S \geq |S| + 1$; but U_S is a space of functions on the finite set $\mathcal{L}$, so $P_\theta(S) = |\mathcal{L}| \geq \dim U_S \geq |S| + 1$ and the main theorem holds. This case covers point

and segment windows, since Z has non-empty interior ($m \geq 2$ and $v_1 \nparallel v_2$). From now on we assume $R_Z(S) \neq \emptyset$.

3. A APPLIED TO THE COLOUR FIELD YIELDS A DOUBLY PERIODIC BACKGROUND

3.1. Realised sectors and adjacent sectors. The complement $\mathbb{R}^2 \setminus \bigcup_i \mathbb{R}v_i$ has exactly $2m$ connected components, open cones: order the $2m$ rays $\{\sigma v_i : \sigma = \pm, i \in [m]\}$ by angle; the components are the *sectors* between consecutive rays. On each sector every π_i has constant sign; write $\epsilon(K) \in \{+, -\}^m$ for the sign vector of the sector K , and call these $2m$ sign vectors *realised*. (For $m \geq 3$ the remaining $2^m - 2m$ sign vectors are not realised.) For a realised ϵ put

$$\Theta_\epsilon := \sum_i \epsilon_i^{\epsilon_i}, \quad \epsilon_i^+ = R_i, \quad \epsilon_i^- = L_i,$$

which is doubly periodic and preserved by every H_j , as is its colour field $e_{\Theta_\epsilon} = (\mathbb{K}[\Theta_\epsilon = a])_a$.

Lemma 3.0 (Sectors adjacent along a ray). *Let the ray σv_i be the common boundary ray of two sectors K, K' . Then $\epsilon(K)$ and $\epsilon(K')$ differ exactly in the coordinate i , and for $j \neq i$ one has $\epsilon(K)_j = \epsilon(K')_j = \text{sign}(\sigma \pi_j(v_i))$. Consequently*

$$\{\Theta_{\epsilon(K)}, \Theta_{\epsilon(K')}\} = \{G_i^{\sigma,L}, G_i^{\sigma,R}\}.$$

In the angular order, the $2m$ sectors form a $2m$ -cycle under the relation of sharing a boundary ray.

Proof. The ray σv_i lies on $\mathbb{R}v_i$ and on no $\mathbb{R}v_j$ with $j \neq i$, the directions being pairwise non-parallel. A small circular arc centred at σv_i crosses only $\mathbb{R}v_i$, so the sign vectors of the two sectors on either side differ only in coordinate i ; for $j \neq i$, the sign of π_j is constant on the arc and equals $\text{sign } \pi_j(\sigma v_i) = \text{sign}(\sigma \pi_j(v_i))$. By (2.0), $\epsilon_j^{\epsilon(K)_j} = \epsilon_j^\sigma$ is precisely the tail choice made in B_i^σ , so $\Theta_{\epsilon(K)}$ and $\Theta_{\epsilon(K')}$ are $R_i + B_i^\sigma$ and $L_i + B_i^\sigma$ in the order dictated by ϵ_i . The cycle structure comes from the angular ordering. $\square$

Lemma 3.1. *For every ray σv_i ,*

$$A(T) e_{G_i^{\sigma,R}} = A(T) e_{G_i^{\sigma,L}}.$$

Consequently $A(T)e_{\Theta_\epsilon}$ takes one and the same value for all realised ϵ ; write

$$b = (b_a)_{a \in \mathbb{F}_p} := A(T)e_{\Theta_\epsilon}.$$

This b is doubly periodic and preserved by every H_j .

Proof. Colour by colour,

$$\mathbb{K}[G_i^{\sigma,R} = a] - \mathbb{K}[G_i^{\sigma,L} = a] = d_{i,\sigma,L,a} - d_{i,\sigma,R,a},$$

which is killed by $A_i(T^{v_i})$ by (P3), hence by A . By Lemma 3.0 the backgrounds of two sectors adjacent along a ray are exactly $G_i^{\sigma,R}$ and $G_i^{\sigma,L}$, so their values of Ae agree; and the $2m$ sectors form a cycle under this adjacency, hence a connected graph, so Ae_{Θ_ϵ} is the same for all realised sectors. Double periodicity and H_j -invariance follow from the corresponding properties of Θ_ϵ together with the fact that A commutes with translations. $\square$

Remark 3.1' (The lemma does not extend to unrealised tail choices). The generalisation “ $Ae_{\Theta_\epsilon} = Ae_{\Theta_{\epsilon'}}$ for all $\epsilon, \epsilon' \in \{L, R\}^m$, realised or not” is *false*. The reason is that Λ_i is defined using only the backgrounds $B_i^\pm$, whereas a mixed tail choice B may have periods along v_i that $B_i^\pm$ do not have, so the spectrum of $\mathbb{K}[R_i + B = a] - \mathbb{K}[L_i + B = a]$ can fall outside Λ_i . An explicit counterexample is in Appendix C.3. Lemma 3.2 uses only realised sectors, so the main line is unaffected.

3.2. $Ae_\theta - b$ has finite support.

Lemma 3.2. *$Ae_\theta - b$ has finite support.*

Proof. Write $M_A = \text{supp}(A)$. Choose $c_i \geq \max(|\ell_i|, |r_i|)$ so that $\{\pi_i < -c_i\} \subseteq \{\pi_i < \ell_i\}$ and $\{\pi_i > c_i\} \subseteq \{\pi_i > r_i\}$, and put

$$\rho_i = c_i + \max_{s \in M_A} |\pi_i(s)|, \quad \Sigma_i = \{z : |\pi_i(z)| \leq \rho_i\}.$$

Call i *active at z* if $z \in \Sigma_i$; otherwise all of $z + M_A$ lies in one and the same tail region of component i , the one on the side $\text{sign } \pi_i(z)$.

(a) *No active direction.* Then $|\pi_i(z)| > \rho_i > 0$ for every i ; put $\epsilon_i = \text{sign } \pi_i(z)$. Regarded as a point of $\mathbb{R}^2$, z lies on no $\mathbb{R}v_i$ and has sign vector ϵ , so ϵ is realised; and $\theta = \Theta_\epsilon$ on $z + M_A$. Hence $(Ae_\theta)(z) = (Ae_{\Theta_\epsilon})(z) = b(z)$.

(b) *At least two active directions.* Then $z \in \Sigma_i \cap \Sigma_j$ with $i \neq j$, an intersection of two strips of bounded width in non-parallel directions, which is bounded and so contains only finitely many lattice points.

(c) *Exactly one active direction i .* Write $z = tv_i + qu_i$ in the basis (v_i, u_i) , so $\pi_i(z) = q$ and $|q| \leq \rho_i$. For $j \neq i$,

$$\pi_j(z) = t \pi_j(v_i) + q \pi_j(u_i), \quad \pi_j(v_i) = \det(v_j, v_i) \neq 0.$$

If

$$|t| > \frac{\rho_i \max_{j \neq i} |\pi_j(u_i)| + \max_{j \neq i} \rho_j}{\min_{j \neq i} |\pi_j(v_i)|}, \quad (3.1)$$

then for every $j \neq i$ and every $s \in M_A$,

$$|\pi_j(z + s)| \geq |t| |\pi_j(v_i)| - |q| |\pi_j(u_i)| - |\pi_j(s)| > \rho_j - |\pi_j(s)| \geq \rho_j - \max_{s' \in M_A} |\pi_j(s')| = c_j,$$

and, because $|t| |\pi_j(v_i)| > |q| |\pi_j(u_i)| + |\pi_j(s)|$, also $\text{sign } \pi_j(z + s) = \text{sign}(t \pi_j(v_i))$. So all of $z + M_A$ lies in the tail region of component j determined by $\text{sign}(t \pi_j(v_i))$. By (2.0) these are exactly the tail choices made in $B_i^{\text{sign}(t)}$, so on $z + M_A$

$$\theta = F_i + B_i^{\text{sign}(t)} = \Xi_i^\sigma, \quad \sigma = \text{sign}(t),$$

whence $(Ae_\theta)(z) = (Ae_{\Xi_i^\sigma})(z)$. By (P3), $A_i(T^{v_i})$ kills $e_{\Xi_i^\sigma} - e_{G_i^{\sigma,R}} = (d_{i,\sigma,R,a})_a$, so

$$(Ae_{\Xi_i^\sigma})(z) = (Ae_{G_i^{\sigma,R}})(z) = b(z),$$

the last step because $G_i^{\sigma,R}$ is the background of one of the realised sectors adjacent along the ray σv_i (Lemma 3.0).

Only finitely many t violate (3.1), and these combine with $|q| \leq \rho_i$ to give finitely many lattice points. Combining (a), (b), (c), $\text{supp}(Ae_\theta - b)$ is finite. $\square$

3.3. Conclusion.

Proposition 3.3. *$A(T)e_\theta = b$. In particular*

$$A\eta = b_\eta := \sum_a w(a)b_a$$

is a doubly periodic field preserved by every H_i .

Proof. By (1.1), $De_\theta = 0$; and b is preserved by every H_i , so $Db = 0$. Hence $D(Ae_\theta - b) = ADe_\theta - Db = 0$. By Lemma 3.2 this function has finite support, so Lemma 1.2 (with $D \neq 0$) forces it to vanish. $\square$

4. AT LEAST ONE TWO-POINT DIFFERENCE IS NON-ZERO

Put $Q_i = \prod_{j \neq i} (T^{H_j} - 1)$ as in (0.2), and $f_i = Q_i \eta$. From $D = (T^{H_i} - 1)Q_i$ and $D\eta = 0$ it follows that f_i has period H_i .

Lemma 4.1. *The support of f_i is contained in a strip of bounded width in the direction v_i , and $f_i \neq 0$.*

Proof. (a) *A global identity.* We have $f_i = T^{nH_i} f_i$ for every n . Let $n \rightarrow +\infty$: on every fixed finite set $T^{nH_i} \theta \rightarrow \Xi_i^+$, the convergence being eventual pointwise stabilisation, so $T^{nH_i} \eta \rightarrow w(\Xi_i^+)$ and therefore

$$f_i = Q_i w(\Xi_i^+). \quad (4.1)$$

Each factor $T^{H_j} - 1$ with $j \neq i$ annihilates $w(G_i^{+,R})$, since H_j preserves that pure-tail field, so

$$f_i = Q_i \delta_i, \quad \delta_i := w(\Xi_i^+) - w(G_i^{+,R}). \quad (4.2)$$

Using $G_i^{+,L}$ instead gives $f_i = Q_i \delta'_i$ with $\delta'_i = w(\Xi_i^+) - w(G_i^{+,L})$.

(b) *A strip of bounded width.* By (P1), $\delta_i \equiv 0$ on $\{\pi_i > r_i\}$ and $\delta'_i \equiv 0$ on $\{\pi_i < \ell_i\}$. Put $c = \max_{s \in \mathcal{E}_i} |\pi_i(s)|$. By (4.2) and (0.2), $f_i(z) \neq 0$ forces $\delta_i(z + s) \neq 0$ for some $s \in \mathcal{E}_i$, whence $\pi_i(z) \leq r_i + c$; the other identity gives $\pi_i(z) \geq \ell_i - c$. So $\text{supp}(f_i) \subseteq \{\ell_i - c \leq \pi_i \leq r_i + c\}$.

(c) *Non-vanishing.* First, $\delta_i \neq 0$: otherwise $w(\Xi_i^+) = w(G_i^{+,R})$, and injectivity of w gives $\Xi_i^+ = G_i^{+,R}$; subtracting B_i^+ gives $F_i = R_i$, which is doubly periodic, contradicting (S2).

Since δ_i vanishes on $\{\pi_i > r_i\}$ but is not identically zero,

$$\pi^* = \max\{\pi_i(z) : \delta_i(z) \neq 0\} (\leq r_i)$$

exists. Choose z^* with $\delta_i(z^*) \neq 0$ and $\pi_i(z^*) = \pi^*$.

Expand Q_i formally as in (0.2). We have $\pi_i(H_C) = \sum_{j \in C} \pi_i(H_j)$ with every $\pi_i(H_j) \neq 0$ for $j \neq i$. Put

$$C_{\min} = \{j \neq i : \pi_i(H_j) < 0\}.$$

For any $C \subseteq [m] \setminus \{i\}$ with $C \neq C_{\min}$,

$$\pi_i(H_C) - \pi_i(H_{C_{\min}}) = \sum_{j \in C \setminus C_{\min}} \pi_i(H_j) - \sum_{j \in C_{\min} \setminus C} \pi_i(H_j) > 0,$$

because the terms of the first sum are positive, those of the second are negative, and at least one of the two sums is non-empty. So $\pi_i(H_{C_{\min}})$ is the *unique* minimum. This is a statement about the π_i -projections and is unaffected by any coincidences among the points H_C themselves.

Evaluate at $z = z^* - H_{C_{\min}}$:

$$f_i(z) = \sum_C (-1)^{m-1-|C|} \delta_i(z + H_C).$$

The term $C = C_{\min}$ contributes $\pm \delta_i(z^*) \neq 0$; for $C \neq C_{\min}$,

$$\pi_i(z + H_C) = \pi^* + (\pi_i(H_C) - \pi_i(H_{C_{\min}})) > \pi^*,$$

so δ_i vanishes there. Hence $f_i(z) = \pm \delta_i(z^*) \neq 0$. $\square$

Lemma 4.2. *For every d the function $J(d, \cdot) := D_z[\eta(z)\eta(z+d)]$ has finite support; and there exists d with $J(d, \cdot) \neq 0$.*

Proof. Finite support: $z \mapsto \eta(z)\eta(z+d)$ is a local function of θ , with window $W = \{0, d\}$; apply Lemma 1.1.

Non-vanishing: choose $i \neq j$, possible since $m \geq 2$. By Lemma 4.1 choose x_i, x_j with $f_i(x_i) \neq 0$ and $f_j(x_j) \neq 0$. Put $d_0 = x_j - x_i$ and

$$C(z) = f_i(z) f_j(z + d_0).$$

Then $C(x_i) = f_i(x_i)f_j(x_j) \neq 0$. Now $\text{supp}(f_i)$ lies in a strip of bounded width in the direction v_i and $\text{supp}(f_j(\cdot + d_0))$ in one of direction v_j , with $v_i \not\parallel v_j$, so $\text{supp}(C)$ is bounded. By Lemma 1.2, $DC \neq 0$.

On the other hand, expanding formally by (0.2),

$$C(z) = \left(\sum_{C_1} \pm \eta(z + H_{C_1}) \right) \left(\sum_{C_2} \pm \eta(z + d_0 + H_{C_2}) \right) = \sum_{C_1, C_2} \pm \eta(z + u_1) \eta(z + u_2),$$

with $u_1 = H_{C_1}$ and $u_2 = d_0 + H_{C_2}$. By translation invariance of D ,

$$DC = \sum_{C_1, C_2} \pm T^{u_1} \left(D_z [\eta(z) \eta(z + u_2 - u_1)] \right) = \sum_{C_1, C_2} \pm T^{u_1} J(u_2 - u_1, \cdot).$$

If every $J(d, \cdot)$ vanished, then $DC = 0$, a contradiction. $\square$

5. THE BI-RECURSION AND THE QUOTIENT ALGEBRA

5.1. The bi-recursion. Regard J as a function of the pair (d, z) : $J(d, z) = D_z [\eta(z) \eta(z + d)]$. Write $T_d^s J(d, z) = J(d + s, z)$ and $T_z^s J(d, z) = J(d, z + s)$.

Lemma 5.1.

$$A(T_d) J = 0, \quad A(T_z T_d^{-1}) J = 0. \quad (5.1)$$

Proof. First identity. For any Laurent polynomial $g = \sum_s g_s T^s$,

$$g(T_d) J(d, z) = \sum_s g_s D_z [\eta(z) \eta(z + d + s)] = D_z [\eta(z) (g(T) \eta)(z + d)].$$

Take $g = A$; by Proposition 3.3, $(A\eta)(z + d) = b_\eta(z + d)$. Expanding D_z by (0.1),

$$D_z [\eta(z) b_\eta(z + d)] = \sum_C (-1)^{m-|C|} \eta(z + H_C) b_\eta(z + H_C + d).$$

Each H_C is a sum of vectors H_i and so preserves b_η ; therefore $b_\eta(z + H_C + d) = b_\eta(z + d)$ and

$$= b_\eta(z + d) \sum_C (-1)^{m-|C|} \eta(z + H_C) = b_\eta(z + d) (D\eta)(z) = 0.$$

Second identity. The operator $(T_z T_d^{-1})^s$ sends (d, z) to $(d - s, z + s)$: the first observation point moves, $z \mapsto z + s$, while the second, $z + d$, does not. Hence

$$\begin{aligned} g(T_z T_d^{-1}) J(d, z) &= \sum_s g_s D_{z'} [\eta(z') \eta(z' + d - s)] \Big|_{z'=z+s} \\ &= \sum_C (-1)^{m-|C|} \sum_s g_s \eta(z + s + H_C) \eta(z + H_C + d) \\ &= \sum_C (-1)^{m-|C|} (g(T) \eta)(z + H_C) \eta(z + H_C + d). \end{aligned}$$

Take $g = A$: then $(A\eta)(z + H_C) = b_\eta(z + H_C) = b_\eta(z)$, so the sum equals

$$b_\eta(z) \sum_C (-1)^{m-|C|} \eta(z + H_C + d) = b_\eta(z) \cdot (D\eta)(z + d) = 0. \quad \square$$

5.2. The quotient algebra. Let $X = (X_1, X_2)$ be independent indeterminates, $K = \mathbb{C}(X_1, X_2)$, and $R = K[Y_1^{\pm 1}, Y_2^{\pm 1}]$. For $u \in \mathbb{Z}^2$ write $Y^u = Y_1^{u_1} Y_2^{u_2}$ and $X^u = X_1^{u_1} X_2^{u_2}$; the monomials $\{Y^u\}_{u \in \mathbb{Z}^2}$ are a K -basis of R . Put

$$a_i = A_i(Y^{v_i}), \quad c_i = A_i(X^{v_i} Y^{-v_i}), \quad a = \prod_i a_i = A(Y), \quad c = \prod_i c_i = A(X/Y).$$

Lemma 5.2. *As a K -vector space, $R/(a, c)$ is spanned by the images of $\{Y^d : d \in (Z - Z) \cap \mathbb{Z}^2\}$.*

Proof. Step 1: no common zeros, and the Chinese remainder theorem. Extend to $\bar{K}$ and test in $(\bar{K}^\times)^2$. Note that $\mathbb{C}$ is algebraically closed, so an element of $\bar{K}$ that is algebraic over $\mathbb{C}$ lies in $\mathbb{C}$, whereas X^u with $u \neq 0$ is transcendental over $\mathbb{C}$.

(i) a_i and c_i have no common zero. From $a_i = 0$ we get $Y^{v_i} = \lambda \in \Lambda_i$; from $c_i = 0$ we get $X^{v_i} Y^{-v_i} = \mu \in \Lambda_i$. Multiplying, $X^{v_i} = \lambda \mu \in \mathbb{C}$, a contradiction.

(ii) If $a_i = a_k = 0$ with $i \neq k$, then every $c_j \neq 0$. From $Y^{v_i} = \lambda$, $Y^{v_k} = \lambda'$ and $\Delta = \det(v_i, v_k) \neq 0$, the elements Y_1^Δ and Y_2^Δ are products of integer powers of λ, λ' , so Y_1, Y_2 are algebraic over $\mathbb{C}$ and hence lie in $\mathbb{C}^\times$. If some $c_j = 0$ then $X^{v_j} = \mu Y^{v_j} \in \mathbb{C}$, a contradiction.

(iii) If $c_j = c_l = 0$ with $j \neq l$, then every $a_i \neq 0$. Put $W = X/Y$, that is $W_1 = X_1/Y_1$ and $W_2 = X_2/Y_2$. From $W^{v_j} = \mu$ and $W^{v_l} = \mu'$ we get, as in (ii), $W_1, W_2 \in \mathbb{C}^\times$, that is $Y_1 = \alpha X_1$ and $Y_2 = \beta X_2$ with $\alpha, \beta \in \mathbb{C}^\times$. If some $a_i = 0$ then $Y^{v_i} = \lambda$, that is $\alpha^{v_{i1}} \beta^{v_{i2}} X^{v_i} = \lambda$ and $X^{v_i} \in \mathbb{C}$, a contradiction.

By (ii), for $i \neq k$ the ideal (a_i, a_k, c) has no zero in $(\bar{K}^\times)^2$, so it equals (1) in $\bar{K}[Y^\pm]$ by the Nullstellensatz, and descends to R by faithful flatness of $R \rightarrow \bar{K}[Y^\pm]$: $(a_i, a_k, c) = R$. That is, the a_i are pairwise coprime in $R/(c)$. By the Chinese remainder theorem,

$$R/(a, c) = R/(\prod_i a_i, c) \simeq \prod_i R/(a_i, c).$$

In $R/(a_i)$: by (i), $(a_i, c_i) = R$, so c_i is a unit; by (iii), for $j \neq l$ with $j, l \neq i$ one has $(a_i, c_j, c_l) = R$, so the c_j are pairwise coprime in $R/(a_i)$. A second application of the Chinese remainder theorem gives

$$R/(a_i, c) = R/(a_i, \prod_j c_j) \simeq \prod_{j \neq i} R/(a_i, c_j),$$

and combining, with the isomorphism given by the natural projections,

$$R/(a, c) \simeq \prod_{i \neq j} R/(a_i, c_j). \quad (5.2)$$

Step 2: a monomial spanning set for each factor. Fix $i \neq j$. Put $u = Y^{v_i}$ and $w = Y^{-v_j}$, let $\mathfrak{L} = \mathbb{Z}v_i + \mathbb{Z}(-v_j)$ and $\nu = [\mathbb{Z}^2 : \mathfrak{L}] = |\det(v_i, v_j)|$. Take the ν lattice points $r_1, \dots, r_\nu$ of the half-open fundamental domain

$$\Pi = \{\alpha v_i - \beta v_j : 0 \leq \alpha < 1, 0 \leq \beta < 1\},$$

a complete set of representatives for $\mathbb{Z}^2/\mathfrak{L}$. Then R is free as a $K[u^\pm, w^\pm]$ -module with basis $\{Y^{r_n}\}$.

The ideal (a_i, c_j) is generated by the elements $a_i = A_i(u)$ and $c_j = A_j(X^{v_j} w)$ of the subring $K[u^\pm, w^\pm]$, so

$$R/(a_i, c_j) \simeq \bigoplus_{n=1}^{\nu} Y^{r_n} \cdot K[u^\pm, w^\pm]/(A_i(u), A_j(X^{v_j} w)).$$

Now $A_i(u)$ has degree d_i in u with both extreme coefficients non-zero (A_i is monic and $A_i(0) \neq 0$), so any integer power of u reduces into the K -span of $\{u^0, \dots, u^{d_i-1}\}$; and $A_j(X^{v_j} w)$ has degree

d_j in w with leading coefficient $X^{d_j v_j} \neq 0$ and constant term $A_j(0) \neq 0$, so any integer power of w reduces into the span of $\{w^0, \dots, w^{d_j-1}\}$. Hence

$$\left\{ Y^{r_n + \alpha v_i - \beta v_j} : 1 \leq n \leq \nu, 0 \leq \alpha < d_i, 0 \leq \beta < d_j \right\} \quad (5.3)$$

spans $R/(a_i, c_j)$. Since $r_n \in \Pi$ — its v_i -coefficient and its $(-v_j)$ -coefficient both lie in $[0, 1]$ — all the exponents in (5.3) lie in

$$P_{ij} := [0, d_i v_i] + [0, -d_j v_j]. \quad (5.4)$$

Step 3: projection onto a factor. Put

$$E_{ij} = \prod_{k \neq i} a_k \prod_{l \neq j} c_l.$$

In the decomposition (5.2):

- on a factor (i', j') with $i' \neq i$: the factor $a_{i'}$ occurs in E_{ij} and is $\equiv 0$ there, so $E_{ij} \equiv 0$;
- on a factor (i, j') with $j' \neq j$: the factor $c_{j'}$ occurs and is $\equiv 0$, so $E_{ij} \equiv 0$;
- on the factor (i, j) : by (i), (ii), (iii), the ideals (a_i, c_j, a_k) for $k \neq i$ and (a_i, c_j, c_l) for $l \neq j$ have no zero in $(\overline{K}^\times)^2$, hence equal R by the Nullstellensatz and faithfully flat descent. Now $(a_i, c_j, x) = R$ means precisely that there is $w \in R$ with $wx \equiv 1 \pmod{(a_i, c_j)}$, so each a_k and each c_l is a unit in $R/(a_i, c_j)$, and so is their product E_{ij} .

Therefore the image of $E_{ij} \cdot (5.3)$ spans the factor (i, j) and vanishes on all the others. Every monomial exponent occurring in $E_{ij} \cdot Y^e$ lies in $\text{Newt}(E_{ij}) + e$, and

$$\begin{aligned} \text{Newt}(E_{ij}) + P_{ij} &\subseteq \sum_{k \neq i} [0, d_k v_k] + \sum_{l \neq j} [0, -d_l v_l] + P_{ij} \\ &= \sum_k [0, d_k v_k] + \sum_l [0, -d_l v_l] = Z + (-Z) = Z - Z, \end{aligned}$$

using $\text{Newt}(a_k) = [0, d_k v_k]$, $\text{Newt}(c_l) = [0, -d_l v_l]$ and the fact that the Newton polygon of a product is the Minkowski sum. Taking the union over all $i \neq j$ gives the spanning statement. $\square$

5.3. The witness lies in $Z - Z$.

Proposition 5.3. *There is $d \in (Z - Z) \cap \mathbb{Z}^2$ with $d \neq 0$ and $J(d, \cdot) \neq 0$.*

Proof. By Lemma 4.2 each $J(d, \cdot)$ has finite support, so

$$\hat{J}(d; X) = \sum_z J(d, z) X^{-z} \in \mathbb{C}[X_1^\pm, X_2^\pm] \subset K$$

is well defined, the supports for different d being allowed to differ. Define a K -linear functional $L: R \rightarrow K$ by setting $L(Y^d) = \hat{J}(d; X)$ on the K -basis and extending K -linearly. That L is K -linear, and not merely $\mathbb{C}$ -linear, is what makes the next step work: coefficients $X^s \in K$ may be pulled out of L .

L kills the ideal (a, c) . Write $A = \sum_s A_s T^s$, so that $a = A(Y) = \sum_s A_s Y^s$ and $c = A(X/Y) = \sum_s A_s X^s Y^{-s}$.

By the first identity of (5.1), for every (d, z) we have $\sum_s A_s J(d + s, z) = 0$. Taking the Laurent transform in z ,

$$\sum_s A_s \hat{J}(d + s; X) = 0 \implies L(Y^d \cdot a) = \sum_s A_s L(Y^{d+s}) = 0.$$

By the second identity of (5.1), for every (d, z) we have $\sum_s A_s J(d - s, z + s) = 0$. Taking the Laurent transform in z and noting

$$\sum_z J(d - s, z + s) X^{-z} = \sum_{z'} J(d - s, z') X^{-z'+s} = X^s \hat{J}(d - s; X),$$

we get $\sum_s A_s X^s \hat{J}(d - s; X) = 0$, that is $L(Y^d \cdot c) = \sum_s A_s X^s L(Y^{d-s}) = 0$.

For an arbitrary $g = \sum_d g_d Y^d \in R$ with $g_d \in K$, K -linearity gives $L(ga) = \sum_d g_d L(Y^d a) = 0$, and likewise $L(gc) = 0$. So L vanishes on (a, c) and descends to $\bar{L}: R/(a, c) \rightarrow K$.

Conclusion. If $J(d, \cdot) = 0$ for every $d \in (Z - Z) \cap \mathbb{Z}^2$, then $\bar{L}$ vanishes on a spanning set supplied by Lemma 5.2, hence $\bar{L} = 0$, hence $L = 0$, hence $\hat{J}(d; X) = 0$ for every d , that is $J \equiv 0$ — contradicting Lemma 4.2.

$d \neq 0$. We have $J(0, z) = D_z[\eta(z)^2]$, and $\eta^2 = w'(\theta)$ with $w'(a) = w(a)^2$ is a single-site colour function, so $D\eta^2 = 0$ by (1.1). $\square$

6. PUTTING TWO POINTS INTO ONE INTEGRAL ZONOTOPE

Lemma 6.1. *Let $Z = \sum_{i=1}^m [0, d_i v_i]$ with integers $d_i \geq 1$, $v_i \in \mathbb{Z}^2$, $m \geq 2$ and $v_1 \nparallel v_2$. Then*

$$(Z - Z) \cap \mathbb{Z}^2 = (Z \cap \mathbb{Z}^2) - (Z \cap \mathbb{Z}^2). \quad (6.1)$$

Proof. The inclusion “ $\supseteq$ ” is clear.

“ $\subseteq$ ”: put $c_0 = \sum_i d_i v_i \in \mathbb{Z}^2$. For $x = \sum_i x_i \in Z$ with $x_i \in [0, d_i v_i]$ we have $c_0 - x = \sum_i (d_i v_i - x_i)$ with $d_i v_i - x_i \in [0, d_i v_i]$, so $c_0 - x \in Z$; that is, $c_0 - Z \subseteq Z$. Applying this once more, $Z = c_0 - (c_0 - Z) \subseteq c_0 - Z$, so $c_0 - Z = Z$ and $-Z = Z - c_0$. Hence

$$Z - Z = Z + (-Z) = Z + Z - c_0 = 2Z - c_0.$$

Let $d \in (Z - Z) \cap \mathbb{Z}^2$ and put $x = d + c_0 \in 2Z \cap \mathbb{Z}^2$.

The vertices of Z are subset sums of the vectors $d_i v_i$ and so are lattice points; thus Z is a two-dimensional lattice polygon. Take a *unimodular triangulation* of Z : every two-dimensional lattice polygon can be subdivided into lattice triangles of area $\frac{1}{2}$ whose vertices are lattice points of Z . Since $x/2 \in Z$, it lies in some unimodular triangle $\triangle ABC$ (boundary included), and $(B - A, C - A)$ is a basis of $\mathbb{Z}^2$. Write

$$\frac{1}{2}x = A + \alpha(B - A) + \beta(C - A), \quad \alpha, \beta \geq 0, \alpha + \beta \leq 1.$$

Then $x - 2A = (2\alpha)(B - A) + (2\beta)(C - A)$. The left-hand side is a lattice point and $(B - A, C - A)$ is a lattice basis, so $2\alpha, 2\beta \in \mathbb{Z}_{\geq 0}$ with $2\alpha + 2\beta \leq 2$. The cases are:

$(2\alpha, 2\beta)$	x	decomposition
(0, 0)	$2A$	$A + A$
(1, 0)	$2A + (B - A) = A + B$	$A + B$
(0, 1)	$A + C$	$A + C$
(2, 0)	$2A + 2(B - A) = 2B$	$B + B$
(0, 2)	$2C$	$C + C$
(1, 1)	$2A + (B - A) + (C - A) = B + C$	$B + C$

In every case $x = p + q$ with $p, q \in \{A, B, C\} \subseteq Z \cap \mathbb{Z}^2$.

Finally $d = x - c_0 = p - (c_0 - q)$, and $c_0 - q \in c_0 - Z = Z$ is a lattice point. $\square$

Corollary 6.2. *The vector d of Proposition 5.3 can be written $d = q_1 - q_0$ with $q_0, q_1 \in Z \cap \mathbb{Z}^2$. Putting $q = q_0$, both q and $q + d$ lie in $Z \cap \mathbb{Z}^2$.*

7. QUADRATIC CHARACTERS MAKE UP THE WHOLE LINEAR DEFICIT

Fix d and q as in Proposition 5.3 and Corollary 6.2, and put

$$h(z) = \eta(z) \eta(z + d), \quad J := Dh \neq 0 \text{ of finite support.}$$

Lemma 7.1. *For every $r \in R_Z(S)$ one has $r + q \in S$ and $r + q + d \in S$.*

Proof. We have $r + Z \subseteq \text{Conv}(S)$ and $q, q + d \in Z \cap \mathbb{Z}^2$, so

$$r + q, r + q + d \in (r + Z) \cap \mathbb{Z}^2 \subseteq \text{Conv}(S) \cap \mathbb{Z}^2 = S. \quad \square$$

Consequently, for each $r \in R_Z(S)$ the *quadratic scalar character*

$$\Phi_r: \mathcal{L} \rightarrow \mathbb{Q}, \quad \Phi_r(\gamma) = w(\gamma(r + q)) \cdot w(\gamma(r + q + d))$$

is well defined, and $\Phi_r(\theta|_{u+S}) = \eta(u + r + q) \eta(u + r + q + d) = h(u + r + q)$.

Lemma 7.2. *The family $\{\Phi_r\}_{r \in R_Z(S)}$ is linearly independent in $\{\mathcal{L} \rightarrow \mathbb{Q}\}/U_S$.*

Proof. Suppose not: there are (c_r) , not all zero, and $\psi \in U_S$ with $\sum_r c_r \Phi_r = \psi$ on $\mathcal{L}$. Substituting $\gamma = \theta|_{u+S}$,

$$\sum_{r \in R_Z(S)} c_r h(u + r + q) = c_0 + \sum_{s \in S} \mu_s \eta(u + s) \quad \text{for all } u.$$

Apply D in u to both sides. The right-hand side gives $Dc_0 + \sum_s \mu_s T^s(D\eta) = 0$ by (1.1); the left-hand side gives $(\sum_r c_r T^{r+q})(Dh) = (\sum_r c_r T^{r+q})J$.

So $(\sum_r c_r T^{r+q})J = 0$. The exponents $r + q$ for $r \in R_Z(S)$ are distinct and the (c_r) are not all zero, so that Laurent polynomial is non-zero; and J is non-zero of finite support. This contradicts Lemma 1.2. $\square$

Theorem 7.3 (Theorem T). $P_\theta(S) \geq |S| + 1$.

Proof. The case $R_Z(S) = \emptyset$ is Remark 2.6. Assume $R_Z(S) \neq \emptyset$ and put

$$V = U_S + \text{span}_{\mathbb{Q}}\{\Phi_r : r \in R_Z(S)\} \subseteq \{\mathcal{L} \rightarrow \mathbb{Q}\}.$$

By Lemma 7.2, $\dim V = \dim U_S + |R_Z(S)|$; by Lemma 2.5,

$$\dim V \geq (|S| - |R_Z(S)| + 1) + |R_Z(S)| = |S| + 1.$$

On the other hand V is a space of functions on the finite set $\mathcal{L}$, so $\dim V \leq |\mathcal{L}| = P_\theta(S)$. $\square$

Remark 7.4 (The balance). The zonotope Z enters the proof twice and with opposite signs: it is the *cost* of the affine relations, Lemma 2.5 subtracting $|R_Z(S)|$, and it is the *capacity* of the quadratic witness, Lemma 7.1 supplying exactly $|R_Z(S)|$ positions. The two cancel exactly, independently of the shape of S and of the size of Z . The numerical experiments of Appendix C show that Lemma 2.5 is an equality on every instance of Case B that we generated.

8. REDUCTION TO THE CONVEX NIVAT CONJECTURE

This section starts from a non-periodic configuration ξ of low convex complexity, extracts a star configuration from X_ξ , and contradicts Theorem 7.3. The chain is

$$\begin{aligned} \text{Theorem 8.4 [11]} &\longrightarrow \text{Lemma 8.5 (mod } p) \longrightarrow \text{Theorems 8.6 and 8.7 [2]} \\ &\longrightarrow \text{Proposition 8.9 (first half-plane, this paper)} \end{aligned}$$

$$\longrightarrow \text{Theorem 8.12 (second half-plane, this paper)} \longrightarrow \text{Lemma 8.17} \longrightarrow \text{Theorem 7.3.}$$

Every link is either a result of [11] or [2], or proved here in full.

Theorem 8.1 (Structure input). *Let ξ be a counterexample of minimal order in the sense of Remark 8.3. Then there are a prime p with $\mathcal{A} \subset \llbracket p \rrbracket$, a non-periodic $\vartheta \in X_\xi$, an $m \geq 2$ and $\mathbb{F}_p$ -valued $G_1, \dots, G_m$ with $\vartheta = \sum_i G_i$ over $\mathbb{F}_p$, each G_i having a non-zero period h_i , the h_i pairwise non-parallel and no component doubly periodic, such that for every i :*

- (i) G_i is fully periodic on a half-plane U_i whose boundary is parallel to h_i ;
- (ii) G_i is fully periodic on a half-plane V_i disjoint from U_i .

Part (i) is proved in §§8.2–8.3 (Proposition 8.9 and Corollary 8.10); part (ii) is proved in §8.4 (Theorem 8.12).

8.1. Half-planes, regions and full periodicity.

Definition. A *half-plane* is a set of the form $\mathcal{H} = \{z \in \mathbb{Z}^2 : \langle z, n \rangle > c\}$ or $\mathcal{H} = \{z \in \mathbb{Z}^2 : \langle z, n \rangle \geq c\}$ with $n \in \mathbb{R}^2 \setminus \{0\}$ and $c \in \mathbb{R}$; we call n its normal. For an integer vector g : if $\langle g, n \rangle \geq 0$ then $\mathcal{H} + g \subseteq \mathcal{H}$; if $\langle g, n \rangle > 0$ then $\bigcup_{N \geq 0} (\mathcal{H} - Ng) = \mathbb{Z}^2$.

Call $R \subseteq \mathbb{Z}^2$ a *lattice-convex region* if $R = C \cap \mathbb{Z}^2$ for some closed convex $C \subseteq \mathbb{R}^2$. Writing $C_R := \overline{\text{Conv}}(R)$ for the closed convex hull, one still has $R = C_R \cap \mathbb{Z}^2$, since $R \subseteq C_R \subseteq C$. Let

$$K_R := \{w \in \mathbb{R}^2 : C_R + w \subseteq C_R\}$$

be the recession cone of C_R , a closed convex cone. If $h \in \mathbb{Z}^2$ satisfies $R + h \subseteq R$, then $C_R + h = \overline{\text{Conv}}(R + h) \subseteq C_R$, that is $h \in K_R$. A half-plane is a lattice-convex region, with $K_R = \{w : \langle w, n \rangle \geq 0\}$.

For a non-empty $\mathcal{U} \subseteq \mathbb{Z}^2$ say that $G : \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ is *fully periodic on $\mathcal{U}$* ([5, Definition 2.1]) if there are $\mathbb{R}$ -linearly independent $h, h' \in \mathbb{Z}^2$ with $\mathcal{U} + h \subseteq \mathcal{U}$, $\mathcal{U} + h' \subseteq \mathcal{U}$, $G(z + h) = G(z)$ and $G(z + h') = G(z)$ for all $z \in \mathcal{U}$. The restriction of a doubly periodic field to any half-plane is fully periodic: take two independent vectors of its period lattice pointing into the half-plane. Conversely:

Lemma 8.2 (Global extension). *Let R be a lattice-convex region and let G be fully periodic on R , with vectors h, h' . Then $\dim K_R = 2$ and $g := h + h' \in \text{int } K_R$; for every $z \in \mathbb{Z}^2$ one has $z + Ng \in R$ for all large N ; and there is a doubly periodic field $\tilde{G}$ with $\text{Per}(\tilde{G}) \supseteq \mathbb{Z}h + \mathbb{Z}h'$ and $\tilde{G} = G$ on R .*

Proof. From $R + h \subseteq R$ and $R + h' \subseteq R$ we get $h, h' \in K_R$; they are linearly independent, so the closed sector they span lies in K_R , its interior is non-empty and contains g , whence $g \in \text{int } K_R$ and $\dim K_R = 2$. Pick $x_0 \in C_R$; by the definition of the recession cone $x_0 + K_R \subseteq C_R$. For any z we have $z + Ng - x_0 = N(g + \frac{z - x_0}{N})$, and for N large the vector in brackets lies in a neighbourhood of g contained in K_R ; hence $z + Ng \in C_R \cap \mathbb{Z}^2 = R$.

Step 1. For $z \in R$ and $k \geq 0$: $z + kh \in R$ and $G(z + kh) = G(z)$, by induction on k ; likewise for h' and for g .

Definition. For $z \in \mathbb{Z}^2$ pick $N \geq 0$ with $z + Ng \in R$ and set $\tilde{G}(z) = G(z + Ng)$. This is well defined: if $N \leq N'$ both work, then $G(z + N'g) = G((z + Ng) + (N' - N)g) = G(z + Ng)$ by Step 1. Taking $N = 0$ shows $\tilde{G} = G$ on R .

Periodicity. Given z , pick N with $z + Ng \in R$; then also $z + h + Ng \in R$, so $\tilde{G}(z + h) = G((z + Ng) + h) = G(z + Ng) = \tilde{G}(z)$ by Step 1. The same for h' . $\square$

Half-planes are the special case: there $K_R = \{w : \langle w, n \rangle \geq 0\}$ and g satisfies $\langle g, n \rangle > 0$.

8.2. The results quoted, and counterexamples of minimal order.

Remark 8.3 (Counterexamples of minimal order). For a non-periodic configuration ξ of low convex complexity with $\mathcal{A} \subset \mathbb{Z}_+$, write $\text{ord}(\xi)$ for the length of a $\mathbb{Z}$ -minimal periodic decomposition of ξ ; by [11] a configuration of low complexity has a non-trivial annihilator and hence a finite periodic decomposition, so ord is finite. If the convex Nivat conjecture has a counterexample, then one may take a counterexample ξ with $\text{ord}(\xi)$ minimal, and for such a ξ every non-periodic $\xi' \in X_\xi$ is again a counterexample, with $\text{ord}(\xi') = \text{ord}(\xi)$.

Proof. Every finite-window pattern of ξ' is a pattern of ξ , so ξ' has low convex complexity; being non-periodic it is a counterexample, and minimality gives $\text{ord}(\xi') \geq \text{ord}(\xi) =: m$. Conversely write $\xi = \sum_{i \leq m} \xi_i$ minimal, with ξ_i of period h_i ; the directions are pairwise distinct, since two components with parallel periods could be merged, contradicting minimality. Then $\varphi = \prod_i (X^{h_i} - 1)$ annihilates ξ ; an annihilation relation is a linear identity on a finite window and therefore passes to the orbit closure, so φ annihilates ξ' , and by [11] the configuration ξ' has a periodic decomposition with at most m terms, that is $\text{ord}(\xi') \leq m$. $\square$

Consequently every result of [2] proved under the standing assumption that the non-periodic configurations of the orbit closure all have the same order ([2, §4.1]) — including [2, Theorem 1.10] and the Claims of its §4 — is available for a counterexample ξ of minimal order; it is under that assumption that Theorem 8.7 below quotes [2, Lemma 4.6].

Theorem 8.4 (Kari–Szabados [11]). *Let $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ with $\mathcal{A} \subset \mathbb{Z}$ finite.*

- (a) *If $P_\xi(S) \leq |S|$ for some non-empty finite $S \subset \mathbb{Z}^2$, then $\text{Ann}_{\mathbb{Z}}(\xi) \neq 0$.*
- (b) *If $\text{Ann}_{\mathbb{Z}}(\xi) \neq 0$, then there are pairwise non-parallel $h_1, \dots, h_m \in \mathbb{Z}^2 \setminus \{0\}$ such that $\varphi = \prod_{i=1}^m (X^{h_i} - 1)$ annihilates ξ ; and every $\eta \in \mathbb{Z}^{\mathbb{Z}^2}$ annihilated by such a φ can be written $\eta = \sum_i \eta_i$ with $\eta_i \in \mathbb{Z}^{\mathbb{Z}^2}$ of period h_i (the values possibly unbounded).*

Both (a) and (b) are in [11]; the argument for (a) is the same as in the proof of Proposition 1.3.

Lemma 8.5 (Passing to the orbit closure and reducing mod p). *Let ξ be as in Theorem 8.4 with $\mathcal{A} \subset \mathbb{Z}_+$, and let $\varphi = \prod_{i=1}^m (X^{h_i} - 1)$ annihilate ξ . Then for every $\vartheta \in X_\xi$ and every prime $p > \max \mathcal{A}$:*

- (a) *φ annihilates ϑ , so $\vartheta = \sum_i \vartheta_i$ with $\vartheta_i \in \mathbb{Z}^{\mathbb{Z}^2}$ of period h_i ;*
- (b) *putting $\bar{\vartheta}_i := \vartheta_i \bmod p: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ one has $\sum_i \bar{\vartheta}_i = \vartheta$, the values of ϑ being read in $\mathbb{F}_p$ (no colours are merged, since $\mathcal{A} \subset \llbracket p \rrbracket$), and each $\bar{\vartheta}_i$ has period h_i ;*
- (c) *after merging the components whose periods are parallel one may assume the h_i pairwise non-parallel; and if ϑ is non-periodic, at least two components remain.*

Proof. (a) At each z the identity $\varphi\vartheta = 0$ involves only $\vartheta|_{z+\text{supp } \varphi}$, so it is a local function of ϑ ; it vanishes on every translate of ξ , and local functions are continuous for the product topology, hence it vanishes on X_ξ . Now apply Theorem 8.4(b). (b) Reduce pointwise. (c) The sum of two components with parallel periods is again periodic (take a common multiple of the two periods); if only one term were left, ϑ would be periodic. $\square$

Theorem 8.6 (Colle [2, Theorem 1.9]). *Let $\mathcal{A} \subset \mathbb{Z}$ and let ξ have a non-trivial annihilator. If for every line ℓ either $-\ell \notin \text{ONED}(\xi)$ or $\ell \notin \text{ONED}(\xi)$, then ξ is fully periodic. In particular, if ξ is non-periodic there is a line ℓ with $\pm\ell \in \text{ONED}(\xi)$.*

The proof of [2, Theorem 1.9] uses the periodic-limit theorem of Kari and Moutot [9, 10] and the Boyle–Lind theorem [1]; it does not depend on the equal-order assumption of [2, §4.1].

Theorem 8.7 (Colle [2, Lemma 4.6]). *Let ξ be a counterexample of minimal order in the sense of Remark 8.3, so that the equal-order assumption of §4.1 of [2] is in force, and let ℓ be as in Theorem 8.6. Then there are a non-periodic $\vartheta \in X_\xi$ and a lattice-convex region R such that $\vartheta|_R$ is fully periodic. Here R is a (ℓ, ℓ') -region: the set of lattice points of a convex unbounded polygonal region whose boundary consists of two unbounded edges, along ℓ and along ℓ' , both nonexpansive lines for X_ξ , and finitely many bounded edges.*

Theorem 8.7 is the conclusion of [2, Lemma 4.6], restated with “fully periodic” as defined in §8.1; its proof there uses Lemma 3.5, Lemma 4.2, Propositions 2.13–2.14 and Claims 4.7–4.16 of [2], of which Claims 4.11 and 4.16 depend on the equal-order assumption. Only three items of it are used below: ϑ is non-periodic, R is a lattice-convex region, and ϑ is fully periodic on R . The position of the two asymptotic directions of the region is not needed: Proposition 8.9(c) shows that it is determined by the decomposition itself.

8.3. The first half-plane. This subsection proves Theorem 8.1(i). The only tools are Lemma 8.2, the difference operators Q_i of (0.2), and the finite-state argument of step (6) of Lemma D.7 in Appendix D.

Lemma 8.8 (One-sided finite-state lemma). *Let $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ have period kv with v primitive and $k \geq 1$, let (v, u) be a basis of $\mathbb{Z}^2$, so $\det(v, u) = \pm 1$, and define the row coordinate of $z = sv + tu$ by $t = t(z) = \det(v, u) \det(v, z)$. Let $H_1, \dots, H_r \in \mathbb{Z}^2$ with $r \geq 1$ satisfy $\beta_j := t(H_j) \neq 0$, and put $\beta_- := \sum_{\beta_j < 0} \beta_j$ and $\beta_+ := \sum_{\beta_j > 0} \beta_j$. If*

$$\Delta := \prod_{j=1}^r (T^{H_j} - 1) G$$

vanishes on the half-plane $\{t \geq c\}$, then there is $q \geq 1$ with $G(z + qu) = G(z)$ for all z with $t(z) \geq c + \beta_-$; in particular G is fully periodic on $\{t \geq c + \beta_-\}$, with the periods kv and qu , neither of which leaves that half-plane. If $\Delta \equiv 0$, then G is doubly periodic.

Proof. Write $z = sv + tu$. Let $\mathcal{W} := \{f: \mathbb{Z} \rightarrow \mathbb{F}_p : f(s+k) = f(s)\}$, a set of cardinality p^k , and for each row t let $G_t \in \mathcal{W}$ be $G_t(s) := G(sv + tu)$. Write $H_j = \alpha_j v + \beta_j u$. Then

$$(T^{H_j} G)(sv + tu) = G((s + \alpha_j)v + (t + \beta_j)u) = (\rho^{\alpha_j} G_{t+\beta_j})(s),$$

where $(\rho f)(s) := f(s+1)$ is an invertible linear map of $\mathcal{W}$. Since ρ commutes with the translations in the t direction, for every t

$$\Delta_t = \sum_{C \subseteq [r]} (-1)^{r-|C|} \rho^{\alpha_C} G_{t+\beta_C}, \quad \alpha_C := \sum_{j \in C} \alpha_j, \quad \beta_C := \sum_{j \in C} \beta_j.$$

The maximum $\beta_+ = \max_C \beta_C$ is attained only at $C_+ := \{j : \beta_j > 0\}$ and the minimum $\beta_- = \min_C \beta_C$ only at $C_- := \{j : \beta_j < 0\}$; put $L := \beta_+ - \beta_- \geq 1$. From $\Delta_t = 0$ for $t \geq c$ we get:

(F) the forward recursion

$$G_{t+\beta_+} = -\rho^{-\alpha_{C_+}} \sum_{C \neq C_+} (-1)^{r-|C|} \rho^{\alpha_C} G_{t+\beta_C},$$

whose right-hand side involves only $G_{t+\beta_-}, \dots, G_{t+\beta_+-1}$;

(B) likewise $G_{t+\beta_-}$ is determined by $G_{t+\beta_-+1}, \dots, G_{t+\beta_+}$.

Let the state be $\mathbf{x}_t := (G_{t+\beta_-}, \dots, G_{t+\beta_+-1}) \in \mathcal{W}^L$ and let $\Sigma := \{\mathbf{x}_t : t \geq c\}$, a finite set. By (F), $\mathbf{x}_{t+1}$ is determined by $\mathbf{x}_t$, so $\Psi(\mathbf{x}_t) := \mathbf{x}_{t+1}$ is a well-defined map $\Sigma \rightarrow \Sigma$; by (B) it is injective on Σ . An injective self-map of a finite set is a bijection, so $(\mathbf{x}_t)_{t \geq c}$ is purely periodic: there is $q \geq 1$ with $\mathbf{x}_{t+q} = \mathbf{x}_t$ for all $t \geq c$. Reading off the first component, $G_{t+q} = G_t$ for $t \geq c + \beta_-$,

that is $G(z + qu) = G(z)$ whenever $t(z) \geq c + \beta_-$. If $\Delta \equiv 0$ the same holds for every t , and G has the two independent periods qu and $. $\square$$

Proposition 8.9 (The first half-plane). *Let $\vartheta : \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ with $\vartheta = \sum_{i=1}^m \vartheta_i$, $m \geq 2$, where ϑ_i has period $h_i = k_i v_i$ with v_i primitive and the v_i pairwise non-parallel. Let R be a lattice-convex region on which ϑ is fully periodic, with vectors a, b , and put $K := K_R$. Then*

- (a) $a, b \in K$ and $\dim K = 2$;
- (b) for each i exactly one of the following holds:
 - (α) the line $\mathbb{R}v_i$ meets $\text{int } K$, and then ϑ_i is doubly periodic;
 - (β), (γ) otherwise there are $\sigma_i \in \{+, -\}$ and $t_i \in \mathbb{Z}$ such that ϑ_i is fully periodic on the half-plane $U_i := \{\sigma_i \pi_i \geq t_i\}$, and U_i contains a translate $R + w_i$ of R ;
- (c) if ϑ is non-periodic and no component is doubly periodic, then K is the sector of opening $< \pi$ spanned by two rays $\sigma_a v_a, \sigma_b v_b$ with $a \neq b$, and its interior contains no $\pm v_j$. In particular K cannot be a half-plane.

Proof. Step 1 (global extension). By Lemma 8.2, $a, b \in K$, $\dim K = 2$, $g := a + b \in \text{int } K$, and there is a doubly periodic $\tilde{\vartheta}$ with $\tilde{\vartheta} = \vartheta$ on R and $\text{Per}(\tilde{\vartheta})$ of finite index. Choose $N_0 \geq 1$ with $N_0 \mathbb{Z}^2 \subseteq \text{Per}(\tilde{\vartheta})$ and put $H_j := N_0 h_j$. Pick $x_0 \in C_R$, so that $x_0 + K \subseteq C_R$.

Step 2 (differencing). Fix i and put $Q_i := \prod_{j \neq i} (T^{H_j} - 1)$, with formal exponent set $\mathcal{E}_i = \{H_C : C \subseteq [m] \setminus \{i\}\}$ and the expansion (0.2). If $z + \mathcal{E}_i \subseteq R$, then

$$(Q_i \vartheta)(z) = \sum_{C \subseteq [m] \setminus \{i\}} (-1)^{m-1-|C|} \tilde{\vartheta}(z + H_C) = \tilde{\vartheta}(z) \cdot (1 - 1)^{m-1} = 0,$$

because every H_C lies in $\text{Per}(\tilde{\vartheta})$. On the other hand H_j is a period of ϑ_j , so $Q_i \vartheta_j = 0$ for $j \neq i$, and therefore

$$D_i := Q_i \vartheta_i = Q_i \vartheta \quad \text{vanishes on } R'_i := \{z : z + \mathcal{E}_i \subseteq R\}.$$

Put $C' := \bigcap_{e \in \mathcal{E}_i} (C_R - e)$. Then $R'_i = C' \cap \mathbb{Z}^2$, and $C' \supseteq y_0 + K$ where $y_0 := x_0 + N'g$ with N' large: indeed for $x \in y_0 + K$ and $e \in \mathcal{E}_i$ one has $x - x_0 + e = N'g + e + (x - y_0) \in K$, because $N'g + e \in K$.

Step 3 (saturation along v_i). D_i has period h_i , so D_i vanishes on $\text{Sat}_i := \{z : (z + \mathbb{Z}h_i) \cap R'_i \neq \emptyset\}$. There are three cases.

(α) $\mathbb{R}v_i \cap \text{int } K \neq \emptyset$. Then v_i or $-v_i$ lies in $\text{int } K$, so for any z the point $z + nh_i$ enters $y_0 + K$ as $n \rightarrow +\infty$ or as $n \rightarrow -\infty$; hence $\text{Sat}_i = \mathbb{Z}^2$ and $D_i \equiv 0$. Apply Lemma 8.8 in its whole-plane case to the basis (v_i, u_i) , the hypothesis $t(H_j) = N_0 \det(v_i, h_j) \neq 0$ holding because the directions are pairwise non-parallel: ϑ_i is doubly periodic.

(β), (γ) $\mathbb{R}v_i \cap \text{int } K = \emptyset$. A convex set and a line missing its interior can be separated, so K lies in one of the two closed half-planes bounded by $\mathbb{R}v_i$; choose σ_i with $K \subseteq \{\sigma_i \pi_i \geq 0\}$. Since $\dim K = 2$, the functional $\sigma_i \pi_i$ does not vanish identically on K . For $\tau > 0$ the chord $K \cap \{\sigma_i \pi_i = \tau\}$ is either a segment in the direction v_i (case (γ): $K \cap \mathbb{R}v_i = \{0\}$, and the length is $\lambda \tau$ with $\lambda > 0$) or a ray (case (β): v_i is parallel to one of the boundary rays of K). In case (β) take any $c_i > 0$; in case (γ) take $c_i > 0$ with $\lambda c_i \geq |h_i|$. If $\sigma_i \pi_i(z - y_0) \geq c_i$, then

$$(z + \mathbb{R}v_i) \cap (y_0 + K) = y_0 + (K \cap \{\sigma_i \pi_i = \sigma_i \pi_i(z - y_0)\})$$

contains a segment of length $\geq |h_i|$ and hence a point of $z + \mathbb{Z}h_i$, these points being spaced $|h_i|$ apart along that line; so $z \in \text{Sat}_i$. Therefore D_i vanishes on the half-plane $\{\sigma_i \pi_i \geq t\}$ with $t := c_i + \sigma_i \pi_i(y_0)$. Apply Lemma 8.8 to the basis $(v_i, \sigma_i u_i)$, for which $t(z) = \sigma_i \pi_i(z)$ and $t(H_j) = \sigma_i N_0 \det(v_i, h_j) \neq 0$: the field ϑ_i is fully periodic on $U_i := \{\sigma_i \pi_i \geq t_i\}$ with $t_i := t + \beta_-$. Since $y_0 + K \subseteq C'$ and $K \subseteq \{\sigma_i \pi_i \geq 0\}$, the half-plane U_i contains a translate of R .

Step 4 (part (c)). Suppose ϑ is non-periodic and no component is doubly periodic. Then case (α) occurs for no i , that is, no line $\mathbb{R}v_i$ meets $\text{int } K$. Now K is a closed convex cone with $\dim K = 2$. If K were a half-plane, at most one line $\mathbb{R}v_i$ could lie on its boundary and all the others would fall into case (α) , a contradiction; and $K = \mathbb{R}^2$ is excluded in the same way. So K is a sector of opening $< \pi$, each of its two boundary rays lies on one of the lines $\mathbb{R}v_a, \mathbb{R}v_b$ with $a \neq b$ — here $m \geq 2$ and the two boundary rays are not parallel — and its interior contains no $\pm v_j$. $\square$

Corollary 8.10 (Theorem 8.1(i)). *Let ϑ and R be as in Theorem 8.7, let $p > \max \mathcal{A}$ be a prime, and let $\vartheta = \sum_i \bar{v}_i$ be the decomposition of Lemma 8.5, with the h_i pairwise non-parallel. Apply Proposition 8.9 to each component and absorb the components falling into case (α) , which are doubly periodic, into any other component by Operation A of Lemma 8.17; on the half-plane U_k of that component the sum is again the restriction of a doubly periodic field, hence still fully periodic. The resulting decomposition $\vartheta = \sum_{i \in I} G_i$ satisfies: $|I| \geq 2$, since ϑ is non-periodic; no component is doubly periodic; the periods are pairwise non-parallel; and each G_i is fully periodic on a half-plane $U_i = \{\sigma_i \pi_i \geq t_i\}$ whose boundary is parallel to h_i , with $U_i \supseteq R + w_i$. Moreover K_R is, by Proposition 8.9(c), the sector spanned by two adjacent rays $\sigma_a v_a, \sigma_b v_b$.*

Remark 8.11. (1) Proposition 8.9 is the one-sided version of Lemma 4.1 of §4 — the support of $f_i = Q_i \eta$ lies in a strip — and Lemma 8.8 is the general form of step (6) of Lemma D.7. (2) In Proposition 8.9(b) neither the non-periodicity of ϑ nor the minimality of the decomposition is required; components falling into case (α) are simply absorbed.

8.4. The second half-plane. This subsection proves Theorem 8.1(ii), in three steps. Proposition 8.14 uses Theorem D.1 to show that, when ϑ is pushed to the limit along a suitable direction, every subsequential limit on the far side of one component is doubly periodic. Lemma 8.16 shows that this forces the component to agree with a doubly periodic field on a half-plane on that side, the point of the argument being that there are only finitely many such limits. Lemma 8.15 guarantees that the directions required can always be found, in a fixed order, so that an induction produces all of the V_i .

Theorem 8.12 (The second half-plane). *In the notation of Corollary 8.10, for every $i \in I$ there is a half-plane V_i with $V_i \cap U_i = \emptyset$ such that $G_i|_{V_i}$ is fully periodic.*

The proof is at the end of the subsection. By step (γ) of Lemma 8.17, every half-plane of full periodicity of an h_i -periodic field that is not doubly periodic is bounded by a line parallel to h_i ; so such a V_i necessarily lies on the side of U_i opposite to U_i , and shrinking either U_i or V_i destroys nothing.

Lemma 8.13 (Bi-periodicity over $\mathbb{F}_p$). *Let $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ have a non-zero period h , let $d \in \mathbb{Z}^2$ be non-parallel to h , and suppose $g := (T^d - 1)G$ is doubly periodic. Then G is doubly periodic.*

Proof. Choose $M \geq 1$ with $Md \in \text{Per}(g)$. From $T^{Md} - 1 = (\sum_{i=0}^{M-1} T^{id})(T^d - 1)$ we get $g' := (T^{Md} - 1)G = \sum_{i=0}^{M-1} T^{id}g$, and $T^{Md}g' = \sum_i T^{id}T^{Md}g = g'$. Hence

$$(T^{pMd} - 1)G = \left(\sum_{i=0}^{p-1} T^{iMd} \right) g' = p g' = 0,$$

so pMd is a period of G , and it is independent of h . $\square$

Safe directions. In the notation of Corollary 8.10 we may change the signs of the v_j so that $\sigma_j = +$ throughout, that is $U_j = \{\pi_j \geq t_j\}$. For $j \in I$ let $\mathcal{K}_j$ be the set of half-planes on which G_j is so far known to be fully periodic: initially $\mathcal{K}_j = \{U_j\}$, and a half-plane V_j disjoint from U_j

is added to $\mathcal{K}_j$ as soon as it has been produced. For $\mathcal{H} \in \mathcal{K}_j$ write $\tilde{G}_j^{\mathcal{H}}$ for the global extension of $G_j|_{\mathcal{H}}$ furnished by Lemma 8.2. Fix $N \geq 1$ divisible by all the k_j and such that $N\mathbb{Z}^2$ is contained in the period lattice of every $\tilde{G}_j^{\mathcal{H}}$. For a non-zero $d \in N\mathbb{Z}^2$ say that d fixes j if $d \parallel h_j$ — then $d \in \mathbb{Z}k_j v_j$ and $T^d G_j = G_j$ — and that d is safe for j if there is $\mathcal{H} = \{\tau\pi_j \geq t\} \in \mathcal{K}_j$, $\tau \in \{+, -\}$, with $\tau\pi_j(d) > 0$. If d is safe for j then $T^{nd} G_j \rightarrow \tilde{G}_j^{\mathcal{H}}$ pointwise as $n \rightarrow \infty$, the convergence being eventual equality at each point: for a fixed z and large n one has $z + nd \in \mathcal{H}$, whence $G_j(z + nd) = \tilde{G}_j^{\mathcal{H}}(z + nd) = \tilde{G}_j^{\mathcal{H}}(z)$.

Proposition 8.14 (Two-component limits). *Let $b, c \in I$ with $b \neq c$, let $d \in N\mathbb{Z}^2$ fix c , let $\pi_b(d) < 0$, and let d be safe for every $j \in I \setminus \{b, c\}$. Let $n_1 < n_2 < \dots$ be such that $T^{n_k d} G_b$ converges pointwise — such a sequence exists by compactness of $\mathbb{F}_p^{\mathbb{Z}^2}$ — and write G_b^∞ for the limit. Then G_b^∞ is doubly periodic.*

Proof. By the preceding paragraph $T^{n_k d} \vartheta \rightarrow \Xi := G_c + B + G_b^\infty$ pointwise, where $B := \sum_{j \neq b, c} \tilde{G}_j^{\mathcal{H}_j}$ is doubly periodic. Since X_ξ is closed, $\Xi \in X_\xi$, so $P_\Xi(S) \leq P_\vartheta(S) \leq |S|$. Choose $h'_c \in (\text{Per}(B) \cap \mathbb{Z}h_c) \setminus \{0\}$; then $\theta_1 := G_c + B$ has period h'_c and $\theta_2 := G_b^\infty$ has period h_b (a pointwise limit of translates keeps the period), and $h'_c \nparallel h_b$. Theorem D.1 supplies a non-zero period d_0 of $\Xi = \theta_1 + \theta_2$. There are three cases.

(i) d_0 is parallel to neither h_b nor h_c . Then $g := (T^{d_0} - 1)\theta_1 = -(T^{d_0} - 1)\theta_2$ has both h'_c and h_b as periods, hence is doubly periodic; Lemma 8.13, applied to θ_1 with $d_0 \nparallel h'_c$, makes θ_1 doubly periodic, so $G_c = \theta_1 - B$ is doubly periodic, contradicting Corollary 8.10.

(ii) $d_0 \parallel h_b$. Pick $d_1 \in (\mathbb{Z}d_0 \cap \mathbb{Z}h_b) \setminus \{0\}$. From $T^{d_1} \Xi = \Xi$ and $T^{d_1} \theta_2 = \theta_2$ we get $T^{d_1} \theta_1 = \theta_1$; as $d_1 \nparallel h'_c$, θ_1 is doubly periodic and we conclude as in (i).

(iii) $d_0 \parallel h_c$. Pick $d_1 \in (\mathbb{Z}d_0 \cap \mathbb{Z}h'_c) \setminus \{0\}$. From $T^{d_1} \theta_1 = \theta_1$ and $T^{d_1} \Xi = \Xi$ we get $T^{d_1} \theta_2 = \theta_2$; as $d_1 \nparallel h_b$, the field $\theta_2 = G_b^\infty$ is doubly periodic. $\square$

Lemma 8.15 (Available directions). *Let $O \subseteq I$ be the set of components for which no V_j has yet been produced, $O \neq \emptyset$, and let N be chosen as above. Then there are $b \in O$, $c \in I \setminus \{b\}$ and $d \in N\mathbb{Z}^2$ satisfying all the hypotheses of Proposition 8.14.*

Proof. If $O = \{b\}$, take any $c \neq b$ and $d := \tau N h_c$ with $\tau \in \{+, -\}$ chosen so that $\pi_b(\tau h_c) < 0$; this is possible because $h_c \nparallel h_b$, so $\pi_b(h_c) \neq 0$. For $j \neq b, c$ we have $j \notin O$, so $\mathcal{K}_j = \{U_j, V_j\}$ contains a half-plane on each side of π_j , and $\pi_j(d) \neq 0$; hence d is safe for j .

If $|O| \geq 2$, put $K_O := \bigcap_{j \in O} \{\pi_j \geq 0\}$. By Corollary 8.10 the inclusion $U_j \supseteq R + w_j$ gives $K_R \subseteq \{\pi_j \geq 0\}$, so $K_O \supseteq K_R$ and $\dim K_O = 2$. Being the intersection of $|O| \geq 2$ closed half-planes through the origin with pairwise distinct boundary lines $\mathbb{R}v_j$, the cone K_O is a closed sector of opening $< \pi$; its two boundary rays lie on two of the lines $\mathbb{R}v_a, \mathbb{R}v_b$ with $a, b \in O$, $a \neq b$, and $\text{int } K_O \subseteq \{\pi_j > 0\}$ for $j \in O$ contains no $\pm v_j$ with $j \in O$. Write $r \subset \mathbb{R}v_a$ and $r' \subset \mathbb{R}v_b$ for the two boundary rays, oriented so that turning anticlockwise from r to r' sweeps exactly K_O . Among the $2m$ rays $\{\pm v_j : j \in I\}$, let $r_c = \varepsilon_c v_c$ ($c \in I$, $\varepsilon_c \in \{+, -\}$) be the first one met when turning anticlockwise from r' . It exists, and $c \neq b$ with $\angle(r', r_c) < \pi$: the angle from r' anticlockwise to $-r$ equals $\pi - \angle(r, r') < \pi = \angle(r', -r')$, so $-r$ occurs before $-r'$. (The index c may lie in O , and may equal a .) Put $d := \varepsilon_c N h_c$.

We verify the hypotheses. (1) d fixes c . (2) The interior of the sector $Q := \text{cone}(r', r_c)$ contains none of the rays $\pm v_j$, since r_c is the first after r' , and its boundary rays are $r' \subset \mathbb{R}v_b$ and $r_c \subset \mathbb{R}v_c$. For $j \in O \setminus \{b, c\}$ the line $\mathbb{R}v_j$ misses $Q \setminus \{0\}$, so π_j has constant sign on $Q \setminus \{0\}$; as $r' \subset K_O \subseteq \{\pi_j \geq 0\}$ and $r' \not\subset \mathbb{R}v_j$, that sign is positive, and in particular $\pi_j(d) > 0$, so d is safe for j (using $U_j \in \mathcal{K}_j$). (3) Q is adjacent to K_O along r' and crosses the line $\mathbb{R}v_b$, because

$\angle(r', r_c) < \pi$; hence $Q \subseteq \{\pi_b \leq 0\}$, and since $r_c \notin \mathbb{R}v_b$ we get $\pi_b(d) < 0$. (4) For $j \notin O$ with $j \neq c$ we have $\mathcal{K}_j = \{U_j, V_j\}$ and $\pi_j(d) \neq 0$, so d is safe for j . $\square$

Lemma 8.16 (Propagation). *Let $G: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ have period kv with v primitive and $k \geq 1$, let (v, u) be a basis of $\mathbb{Z}^2$ with $\det(v, u) = 1$ and put $\pi := \det(v, \cdot)$. Let $d \in \mathbb{Z}^2$ satisfy $\delta := -\pi(d) > 0$. If every subsequential limit of the sequence $(T^{nd}G)_{n \geq 1}$ is doubly periodic, then there are $\beta \in \mathbb{Z}$ and a doubly periodic field θ_* with $G = \theta_*$ on $\{\pi \leq \beta\}$.*

Proof. Let Ω be the set of limit points of the sequence: $y \in \Omega$ if and only if $T^{n_k d}G \rightarrow y$ pointwise for some $n_1 < n_2 < \dots$. By compactness of $\mathbb{F}_p^{\mathbb{Z}^2}$, $\Omega \neq \emptyset$; it is closed; and $T^d\Omega = T^{-d}\Omega = \Omega$, since replacing the indices n_k by $n_k \pm 1$ does not affect the existence of a subsequential limit. Every element of Ω has period kv , each $T^{nd}G$ having it and pointwise limits keeping it, and is doubly periodic by hypothesis. Call $\{\pi = t\}$ the t -th row and put $B_W := \{sv + tu : |s|, |t| \leq W\}$. Two fields of period kv that agree at k consecutive lattice points of a row agree on the whole row; so agreement on B_W with $W \geq k$ implies agreement on the rows $-W$ to W . For a doubly periodic y of period kv set

$$\text{Tr}(y) := \min\{\pi(\lambda) : \lambda \in \text{Per}(y), \pi(\lambda) > 0\},$$

which is defined because $\text{Per}(y)$ has finite index.

Step 1: Ω is finite. First, $\{y \in \Omega : \text{Tr}(y) \leq P\}$ is finite for every P . Indeed take $\lambda \in \text{Per}(y)$ with $\pi(\lambda) = \text{Tr}(y)$; subtracting a multiple of kv we may assume $\lambda = s'v + \text{Tr}(y)u$ with $0 \leq s' < k$. For any z write $\pi(z) = m \text{Tr}(y) + r$ with $0 \leq r < \text{Tr}(y)$; then $y(z) = y(z - m\lambda)$ and $\pi(z - m\lambda) = r$. So y is determined by λ together with the values of y on $\{sv + tu : 0 \leq s < k, 0 \leq t < \text{Tr}(y)\}$, and there are only finitely many such data.

Suppose Ω were infinite. Then there are $y_n \in \Omega$ with $\text{Tr}(y_n) \rightarrow \infty$; passing to a subsequence, $y_n \rightarrow \theta \in \Omega$. Now θ is doubly periodic; put $P := \text{Tr}(\theta)$ and choose $\varpi \in \text{Per}(\theta)$ with $\pi(\varpi) = P$. From $y_n \rightarrow \theta$ we may choose $W_n \rightarrow \infty$ with $y_n = \theta$ on B_{W_n} , hence, once $W_n \geq k$, on the rows $-W_n$ to W_n .

Call the t -th row ϖ -good for y_n if $y_n(z + \varpi) = y_n(z)$ for every z with $\pi(z) = t$. The rows $-W_n$ to $W_n - P$ are ϖ -good: both z and $z + \varpi$ lie in rows where y_n and θ agree, and $\theta(z + \varpi) = \theta(z)$. Let J_n be the largest integer interval containing $[-W_n, W_n - P]$ and consisting of ϖ -good rows. If $J_n = \mathbb{Z}$ then $\varpi \in \text{Per}(y_n)$ and $\text{Tr}(y_n) \leq P$, contradicting $\text{Tr}(y_n) \rightarrow \infty$ for large n . So $J_n = [a_n, b_n]$ has at least one finite endpoint; passing to a subsequence we may assume a_n finite for every n , the case of finite b_n being symmetric (replace $T^{q_n d}$ below by $T^{-q_n d}$ and $\{\pi \geq \delta\}$ by $\{\pi \leq -\delta\}$). Then $a_n \leq -W_n$ and the row $a_n - 1$ is not ϖ -good: there is z_n with $\pi(z_n) = a_n - 1$ and $y_n(z_n + \varpi) \neq y_n(z_n)$.

Put $q_n := \lceil -a_n/\delta \rceil \geq 1$, so that $r_n := a_n + q_n\delta \in [0, \delta)$, and put $y'_n := T^{q_n d}y_n \in \Omega$. Since $\pi(z + q_n d) = \pi(z) - q_n\delta$, the t -th row is ϖ -good for y'_n exactly when the row $t - q_n\delta$ is ϖ -good for y_n . Hence the rows r_n to $b_n + q_n\delta$ are ϖ -good for y'_n , where $b_n + q_n\delta \geq (W_n - P) + (-a_n) \geq 2W_n - P \rightarrow \infty$, while the row $r_n - 1$ is not: the point $z'_n := z_n - q_n d$ satisfies $\pi(z'_n) = r_n - 1$ and $y'_n(z'_n + \varpi) \neq y'_n(z'_n)$. Using the kv -periodicity, translate z'_n along v into $\{sv + tu : 0 \leq s < k\}$; this yields a point z''_n of the finite set $F := \{sv + tu : 0 \leq s < k, -1 \leq t \leq \delta - 2\}$ with $y'_n(z''_n + \varpi) \neq y'_n(z''_n)$. Passing to a further subsequence we may assume $z''_n = z_*$ and $r_n = r_*$ constant and $y'_n \rightarrow y \in \Omega$, Ω being closed.

Then: (i) $y(z + \varpi) = y(z)$ whenever $\pi(z) \geq \delta$ — for large n one has $r_n \leq \delta - 1 < \pi(z) \leq b_n + q_n\delta$, so the row $\pi(z)$ is ϖ -good for y'_n , and one lets $n \rightarrow \infty$; and (ii) $y(z_* + \varpi) \neq y(z_*)$, the two values lying in the finite set $\mathbb{F}_p$ and converging, so that the inequality survives in the limit.

But $y \in \Omega$ is doubly periodic: taking $\lambda \in \text{Per}(y)$ with $\pi(\lambda) > 0$ and M with $\pi(z_* + M\lambda) \geq \delta$, part (i) gives

$$y(z_* + \varpi) = y(z_* + \varpi + M\lambda) = y(z_* + M\lambda) = y(z_*),$$

contradicting (ii). So Ω is finite.

Step 2: eventual agreement. Put $\Lambda := \bigcap_{y \in \Omega} \text{Per}(y)$, of finite index, and choose $N' \geq 1$ with $N'\mathbb{Z}^2 \subseteq \Lambda$; every element of Ω then has period $N'u$. Put $W := \max(k, N' + \delta)$.

(a) For all large n the field $T^{nd}G$ agrees on B_W with some element of Ω . Otherwise there are $n_1 < n_2 < \dots$ with $T^{n_k d}G|_{B_W}$ outside the finite set $\{y|_{B_W} : y \in \Omega\}$; passing to a convergent subsequence with limit $y \in \Omega$ gives $T^{n_k d}G|_{B_W} = y|_{B_W}$ for large k , a contradiction. For such n choose $y^{(n)} \in \Omega$ with $T^{nd}G = y^{(n)}$ on B_W , hence, $W \geq k$, on the rows $-W$ to W .

(b) Let $n \geq n_0$ be such n . For $\pi(z) \in [-W + \delta, W]$ one has $\pi(z + d) \in [-W, W]$, so

$$y^{(n+1)}(z) = (T^{(n+1)d}G)(z) = (T^{nd}G)(z + d) = y^{(n)}(z + d) = (T^d y^{(n)})(z).$$

Thus $T^d y^{(n)}$ and $y^{(n+1)}$ agree on $2W - \delta + 1 \geq N'$ consecutive entire rows. Both lie in Ω , since $T^d \Omega = \Omega$, and both have period $N'u$; any row can be translated by a multiple of $N'u$ into those N' consecutive rows, so the two fields agree everywhere: $T^d y^{(n)} = y^{(n+1)}$.

(c) Hence $y^{(n)} = T^{(n-n_0)d} y^{(n_0)}$ for $n \geq n_0$. For $\pi(w) \in [-n\delta - W, -n\delta + W]$ one has $\pi(w - nd) \in [-W, W]$, so

$$G(w) = (T^{nd}G)(w - nd) = y^{(n)}(w - nd) = y^{(n_0)}(w - nd + (n - n_0)d) = (T^{-n_0 d} y^{(n_0)})(w).$$

As n runs over the integers $\geq n_0$, the row intervals $[-n\delta - W, -n\delta + W]$ cover $(-\infty, -n_0\delta + W]$, consecutive intervals overlapping because $2W \geq \delta$. Take $\theta_* := T^{-n_0 d} y^{(n_0)}$, which is doubly periodic, and $\beta := -n_0\delta + W$. $\square$

Proof of Theorem 8.12. Induct, starting from $O := I$, the set of components for which no V_j has yet been produced. While $O \neq \emptyset$, choose N as in the paragraph on safe directions — it depends on the current $\mathcal{K}_j$ — and take $b \in O$, c and d as in Lemma 8.15. By Proposition 8.14 every subsequential limit of $(T^{nd}G_b)_{n \geq 1}$ is doubly periodic, so Lemma 8.16, applied to $G := G_b$, $v := v_b$, $u := u_b$ and this d (for which $\pi_b(d) < 0$), yields β and a doubly periodic θ_* with $G_b = \theta_*$ on $\{\pi_b \leq \beta\}$. Put $V_b := \{\pi_b \leq \min(\beta, t_b - 1)\}$. Then $V_b \cap U_b = \emptyset$ and $G_b|_{V_b}$ is fully periodic, with the periods $k_b v_b$ and any $\lambda \in \text{Per}(\theta_*)$ with $\pi_b(\lambda) < 0$, neither of which leaves V_b . Add V_b to $\mathcal{K}_b$ and remove b from O . Each step decreases $|O|$ by one, so after $|I|$ steps $O = \emptyset$ and every component has the required V_i . $\square$

8.5. Normalisation.

Lemma 8.17 (Normalisation). *Let $\theta: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ be non-periodic and suppose $\theta = \sum_{i=1}^m G_i$, where each $G_i: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ has a non-zero period h_i and there are disjoint half-planes U_i, V_i on which G_i is fully periodic. Then θ is a star configuration.*

Call a family $(G_i, h_i, U_i, V_i)_{i=1}^m$ with these properties an *admissible decomposition* of θ , of length m . Since θ is non-periodic, every admissible decomposition has $m \geq 2$: for $m = 1$ we would have $\theta = G_1$ periodic.

Proof. Step (γ): alignment of the components that are not doubly periodic. Let G have a non-zero period h , not be doubly periodic, and be fully periodic on disjoint half-planes U with normal n and V with normal n' . *Claim:* $\langle h, n \rangle = 0$. Let $\tilde{G}$ be the global extension of $G|_U$ from Lemma 8.2 and put $\Gamma_U := \text{Per}(\tilde{G})$, of finite index. Suppose $\langle h, n \rangle \neq 0$; replacing h by $-h$ we may assume $\langle h, n \rangle > 0$. For $z \in U$ we have $z + h \in U$, so $\tilde{G}(z + h) = G(z + h) = G(z) = \tilde{G}(z)$: the Γ_U -periodic function $\tilde{G}(\cdot + h) - \tilde{G}$ vanishes on U . But a Γ_U -periodic function vanishing on a

half-plane vanishes everywhere: Γ_U contains some $N'\mathbb{Z}^2$ and hence a vector g with $\langle g, n \rangle > 0$, and any z is translated into U by $z + Ng$ for N large. So $\tilde{G}$ has period h . Now for any z pick N with $z + Nh \in U$: then $G(z) = G(z + Nh) = \tilde{G}(z + Nh) = \tilde{G}(z)$, that is $G = \tilde{G}$ is doubly periodic, a contradiction.

Hence $n \perp h$. Write $h = kv$ with v primitive and $k \geq 1$, after changing the sign of h if necessary, and put $\pi = \det(v, \cdot)$. Since $v^\perp = (-v_2, v_1)$ satisfies $\langle z, v^\perp \rangle = \pi(z)$ and $n \parallel v^\perp$, we have $\langle z, n \rangle = \kappa \pi(z)$ for some $\kappa \in \mathbb{R}^\times$, so U is either $\{\pi < \ell\}$ or $\{\pi > r\}$ with $\ell, r \in \mathbb{Z}$. The same applies to V . If U and V were both of the type $\{\pi < \cdot\}$, or both of the type $\{\pi > \cdot\}$, they would intersect; so one is $\{\pi < \ell\}$ and the other is $\{\pi > r\}$, and disjointness says exactly $\ell \leq r + 1$. Taking $\tilde{L}, \tilde{R}$ to be the global extensions of G on these two half-planes, G satisfies (S1) with period kv , (S2), and (S3).

Operation A: absorbing a doubly periodic component. Suppose G_i is doubly periodic and $m \geq 2$; pick $k \neq i$. Since $\text{Per}(G_i)$ has finite index, $\text{Per}(G_i) \cap \mathbb{Z}h_k$ contains some $h'_k \neq 0$, and $G'_k := G_k + G_i$ has period h'_k . On U_k we have $G'_k = \tilde{G}_k + G_i$ by Lemma 8.2, the restriction of a doubly periodic field, hence fully periodic; likewise on V_k . Deleting G_i and replacing G_k by G'_k gives an admissible decomposition of length $m - 1$.

Operation C: merging parallel components. Suppose G_i, G_k with $i \neq k$ are both not doubly periodic and $h_i \parallel h_k$. By Step (γ) there is a primitive v with $h_i = \pm k_i v$, $h_k = \pm k_k v$, and $\{U_i, V_i\} = \{\{\pi < \ell_i\}, \{\pi > r_i\}\}$, $\{U_k, V_k\} = \{\{\pi < \ell_k\}, \{\pi > r_k\}\}$, where $\pi = \det(v, \cdot)$. Then $G_i + G_k$ has period $\text{lcm}(k_i, k_k)v$, equals $\tilde{L}_i + \tilde{L}_k$ on $\{\pi < \min(\ell_i, \ell_k)\}$ and $\tilde{R}_i + \tilde{R}_k$ on $\{\pi > \max(r_i, r_k)\}$, both restrictions of doubly periodic fields and hence fully periodic; and these two half-planes are disjoint, since $\min(\ell_i, \ell_k) \leq \ell_i \leq r_i + 1 \leq \max(r_i, r_k) + 1$. Merging gives an admissible decomposition of length $m - 1$.

Termination and conclusion. Starting from the given admissible decomposition, repeat: if some component is doubly periodic, apply Operation A; otherwise, if some pair is parallel, apply Operation C. Each step strictly decreases the length, so the process terminates. At termination no component is doubly periodic, the periods are pairwise non-parallel, and $m \geq 2$. By Step (γ) every component satisfies (S1)–(S3) and the primitive directions v_i are pairwise non-parallel. So θ is a star configuration. $\square$

8.6. The convex Nivat conjecture.

Theorem 8.18 (Convex Nivat conjecture). *Let $\xi: \mathbb{Z}^2 \rightarrow \mathcal{A}$ with $\mathcal{A}$ finite, and suppose there is a non-empty finite lattice-convex S with $P_\xi(S) \leq |S|$. Then ξ is periodic.*

Proof. Suppose the conjecture has a counterexample. Embed the alphabet into $\mathbb{Z}_{>0}$ by an injective encoding, which changes neither the number of patterns nor periodicity, and use Remark 8.3 to choose a counterexample of minimal ord, still written ξ ; it is non-periodic and has low convex complexity.

(0) *The annihilator.* Theorem 8.4 supplies pairwise non-parallel $h_1, \dots, h_m$ and $\varphi = \prod_i (X^{h_i} - 1)$ annihilating ξ ; we may take $m = \text{ord}(\xi)$.

(1) *The region.* Theorem 8.6 supplies ℓ , and Theorem 8.7 a non-periodic $\vartheta \in X_\xi$ together with a lattice-convex region R on which ϑ is fully periodic.

(2) *The decomposition.* Take any prime $p > \max \mathcal{A}$; then Lemma 8.5 gives $\vartheta = \sum_i \bar{\vartheta}_i$ over $\mathbb{F}_p$ with each $\bar{\vartheta}_i$ of period h_i , the directions being pairwise distinct.

(3) *The first half-plane.* Corollary 8.10 gives $\vartheta = \sum_{i \in I} G_i$ with $|I| \geq 2$, no component doubly periodic, pairwise non-parallel periods, and each G_i fully periodic on U_i .

(4) *The second half-plane.* Theorem 8.12 gives half-planes V_i disjoint from U_i with $G_i|_{V_i}$ fully periodic. At this point the whole of Theorem 8.1 holds.

(5) *The complexity does not increase.* Since $\mathcal{A} \subset \llbracket p \rrbracket$, reading the values of ϑ in $\mathbb{F}_p$ merges no colours: as an $\mathbb{F}_p$ -valued configuration, ϑ has the same pattern set and the same periodicity as it has as an $\mathcal{A}$ -valued one. Every finite-window pattern of a configuration in X_ξ is a pattern of ξ , a limit in the product topology agreeing with some translate on any finite window. Hence

$$P_\vartheta(S) \leq P_\xi(S) \leq |S|. \quad (8.1)$$

(6) *Normalisation.* ϑ is non-periodic, $\vartheta = \sum_{i \in I} G_i$ over $\mathbb{F}_p$ with each G_i of period h_i and fully periodic on the disjoint half-planes U_i, V_i . By Lemma 8.17, ϑ is a star configuration. $\square$

(7) *Contradiction.* By Theorem 7.3, $P_\vartheta(S) \geq |S| + 1$, contradicting (8.1). $\square$

Corollary 8.19 (Nivat's conjecture). *If $P_\xi([1, n] \times [1, k]) \leq nk$ for some $n, k \geq 1$, then ξ is periodic. (A rectangle is lattice-convex.)*

APPENDIX A. SKELETON OF THE PROOF

- §1 $\exists a : DI_a \neq 0 \implies$ Theorem 7.3 at once, no convexity needed; otherwise $DI_a \equiv 0$ and $D\eta = 0$.
- §2 $\Lambda_i = \text{exceptional spectrum}, A = \prod_i A_i(T^{v_i}), Z = \text{Newt}(A);$
 $f(T)\eta = \text{const} \implies A \mid f \implies \dim U_S \geq |S| - |R_Z(S)| + 1.$
- §3 Ae agrees on adjacent realised sectors $\implies Ae_\theta = b$, doubly periodic, all H_i -invariant.
- §4 $f_i = Q_i\eta \neq 0$ and transversally bounded $\implies \exists d : J(d, \cdot) = D_z[\eta(z)\eta(z+d)] \neq 0.$
- §5 $A(T_d)J = A(T_z T_d^{-1})J = 0; R/(a, c)$ spanned by $Y^{Z-Z} \implies d \in (Z - Z) \cap \mathbb{Z}^2.$
- §6 $(Z - Z) \cap \mathbb{Z}^2 = (Z \cap \mathbb{Z}^2) - (Z \cap \mathbb{Z}^2) \implies d = q_1 - q_0, q_0, q_1 \in Z.$
- §7 $|R_Z(S)|$ quadratic characters independent modulo $U_S \implies P_\theta(S) \geq |S| + 1.$
- §8 $[11], [2] \xrightarrow{\text{Prop. 8.9}} U_i \xrightarrow{\text{Thm. 8.12}} V_i \xrightarrow{\text{Lem. 8.17}} \text{star configuration}$
 $\xrightarrow{\text{Thm. 7.3}} \text{convex Nivat} \implies \text{Nivat}.$

Dependencies: §§0–7 are self-contained, using only the standard facts that $\mathbb{C}[T^\pm]$ is a unique factorisation domain, Newton-polygon additivity (Ostrowski), the Chinese remainder theorem, the Nullstellensatz, and the existence of unimodular triangulations of lattice polygons. The external inputs to §8 are Kari and Szabados [11] and Colle [2] only: the first supplies the annihilator and the periodic decomposition, the second a line both of whose directions are one-sided nonexpansive ([2, Theorem 1.9]) and a non-periodic configuration in the orbit closure that is fully periodic on a region ([2, Lemma 4.6]). The $\mathbb{F}_p$ and convex form of Szabados' theorem, used by Proposition 8.14, is proved in Appendix D.

APPENDIX B. INDEPENDENT EDGE-INCREMENT RESULTS, NOT PART OF THE PROOF CHAIN

Before the argument of §§0–8 took its present form, the same question was approached along a different route, by comparing $P_\theta(S)$ with $P_\theta(S \setminus E)$ for an edge E of the polygon $\text{Conv}(S)$ and controlling the increment through the combinatorics of the edge directions of $\text{Conv}(S)$ against the directions v_i . That route produced a number of statements of independent interest: an exact identity for the lattice width of $\text{Conv}(S)$ in a given direction in terms of its edge data, a divisibility relation between the determinants of three consecutive edge directions, a propagation principle for sums of functions each of which depends on a single one of the projections π_i

and which vanish on enough consecutive layers, a description of the periods surviving after repeated differencing, a count of the far-field patterns along an edge, and lower bounds of the form $P_\theta(S) - P_\theta(S \setminus E) \geq 2$ for short edges and of the form $P_\theta(S) - P_\theta(S \setminus E) \geq$ (a row-length quantity) in general. None of them is used above, their proofs are not reproduced here, and the notation they require is not set up in this paper; they will be treated separately. Nothing in §§0–8 depends on any of them.

APPENDIX C. NUMERICAL VERIFICATION

The scripts are `star.py` (construction, enumeration and all checks), `verify.py` and `verify2.py` (the two randomised experiments), and `lemma31.py` (the counterexample of §C.3). They depend only on `numpy` and are supplied as ancillary files together with their output for the stated seeds; every number below is read off that output.

C.1. Method.

- *Configurations.* Each component is specified by a primitive direction v_i , a tangential period k_i , an arbitrary value table on the transition strip $[\ell_i, r_i]$, and left and right tails L_i, R_i given, in the basis (v_i, u_i) of §0.3, as doubly periodic tables in $(t \bmod a, s \bmod k_i)$. Axioms (S1) and (S3) hold by construction. Axiom (S2) is decided exactly: F_i is doubly periodic if and only if it coincides with the global extension of its left tail (a doubly periodic function vanishing on a half-plane vanishes identically), and that coincidence is tested on the strip and on one full period beyond it. With M the least common multiple of all the k_j and of all the tail period indices, the lattice Γ of §0.3 contains $M\mathbb{Z}^2$, and the program uses $H_i = Mk_i v_i$; by Remark 1.4 this choice, which need not be the minimal one, does not affect the Case A/Case B decision, and it does not affect Λ_i either, since refining the period only adds zero Fourier coefficients.
- *Pattern counts.* The S -patterns are enumerated at all anchors of the square box of radius $R = 90$ and again of radius $R = 180$. Away from the crossing region of the strips the patterns along each strip are H_i -periodic and, inside a sector, $M\mathbb{Z}^2$ -periodic, so the count is exact as soon as the box is larger than the crossing region plus one period plus $\text{diam}(S)$; the two counts agreed on every window of every configuration reported below.
- *Deciding the case.* DI_a is computed on the box of radius 40 from the formal expansion (0.1); all-zero for every colour a means Case B.
- *Spectrum and zonotope.* The fields $\Xi_i^\pm$ and $G_i^{\sigma, \epsilon}$ are built as in §2.1; for each difference field $d_{i, \sigma, \epsilon, a}$ a length- κ_i DFT is taken row by row over all rows within $2M + 2$ of the strip (outside the strip the rows repeat with period M), and the union of the frequencies with coefficient of modulus above 10^{-9} gives Λ_i, d_i and Z . The set $R_Z(S)$ is decided by testing whether all subset sums of the $d_i v_i$, translated by r , lie in $\text{Conv}(S)$.
- *Ranks.* The encoding w is a random injection into $\{1, \dots, 100\}$. The rank of the $|\mathcal{L}| \times (|S| + 1)$ matrix $(1, w(\gamma(s)))_{\gamma \in \mathcal{L}, s \in S}$ is computed over $\mathbb{F}_q$ with $q = 2^{31} - 1$ and reported as $\dim U_S$; it is a lower bound for the rational rank, which is the direction needed to verify Lemma 2.5. In Case B with $R_Z(S) \neq \emptyset$ the matrix is extended by the columns $\Phi_r(\gamma) = w(\gamma(r + q)) w(\gamma(r + q + d))$, $r \in R_Z(S)$, of §7, and the rank $H_2(S)$ of the extended matrix is compared with $\dim U_S + |R_Z(S)|$ (Lemma 7.2) and with $|S| + 1$ (Theorem 7.3).
- *Witness and background.* In Case B, for every non-zero $d \in (Z - Z) \cap \mathbb{Z}^2$ the function $J(d, \cdot)$ is computed on the box of radius 40 and the non-zero instances are collected (Proposition 5.3); the first of them, with a q as in Corollary 6.2, is the pair (d, q) used

for the Φ_r . The polynomial A is built with complex coefficients from the Λ_i , and Ae_θ is checked to be $M\mathbb{Z}^2$ -periodic on the box (Proposition 3.3).

C.2. Results. Experiment 1 (script `verify.py`, seed 1). We generated 84 random star configurations with $p \in \{2, 3, 5\}$, $m \in \{2, 3\}$, directions drawn without repetition from the 8 primitive vectors $(1, 0), (0, 1), (1, \pm 1), (1, \pm 2), (2, \pm 1)$, $k_i \in \{1, 2, 3\}$, tail periods of 1 or 2 rows, strip widths 1 to 3 with $\ell_i \in \{-1, 0, 1\}$, all tables uniformly random over $\mathbb{F}_p$ subject to (S2); component 1 has independently drawn left and right tails, the other components have $L_i = R_i$. Windows: the 2×3 , 3×3 , 2×4 and 4×2 rectangles, the 6-point triangle with vertices $(0, 0), (2, 0), (0, 2)$, the 8-point hexagon with vertices $(1, 0), (2, 0), (3, 1), (2, 2), (1, 2), (0, 1)$, the 5-point rhombus with vertices $(0, 0), (1, 1), (2, 0), (1, -1)$, and the 3-point segment $\{0, 1, 2\} \times \{0\}$. Result: 81 configurations in Case A and 3 in Case B; $P_\theta(S) \geq |S| + 1$ held on all 8 windows of all 84 configurations. The three Case-B instances have $d_i \in \{3, 6\}$, so Z is larger than every window and $R_Z(S) = \emptyset$ throughout; on them $\dim U_S$ was exactly $|S| + 1$ on every window, the witness d existed (192, 310 and 28 non-zero d respectively), and Ae_θ was doubly periodic to within $2 \cdot 10^{-15}$. These are instances of Remark 2.6; Experiment 2 targets $R_Z(S) \neq \emptyset$.

Experiment 2 (script `verify2.py`, seed 7). We generated 157 configurations of the form $\sum_i w_i \circ \pi_i$, so $k_i = 1$ and constant tails: each $w_i: \mathbb{Z} \rightarrow \mathbb{F}_p$ takes a constant value c_i outside a window of length 1 to 4 and arbitrary values inside it, with $L_i = R_i = c_i$; $p \in \{2, 3, 5\}$, $m = 2$ with probability 0.7 and $m = 3$ with probability 0.3, directions drawn from the 11 primitive vectors $(1, 0), (0, 1), (1, \pm 1), (1, \pm 2), (2, \pm 1), (1, \pm 3), (3, 1)$. Windows: the 3×4 , 4×4 , 5×3 and 4×3 rectangles, the 15-point triangle with vertices $(0, 0), (4, 0), (0, 4)$, the 17-point hexagon with vertices $(1, 0), (3, 0), (4, 2), (3, 4), (1, 4), (0, 2)$, the 13-point rhombus with vertices $(0, 0), (2, 2), (4, 0), (2, -2)$, and the 12-point parallelogram with vertices $(0, 0), (3, 0), (5, 2), (2, 2)$. Result: 128 in Case A and 29 in Case B, all with $d_i = 1$; $P_\theta(S) \geq |S| + 1$ held on all 8 windows of all 157 configurations, and every check listed in §C.1 passed on every Case-B instance. Of the 29 Case-B configurations, 14 have $R_Z(S) \neq \emptyset$ for at least one window, giving 35 configuration-window pairs with $R_Z(S) \neq \emptyset$ and $|R_Z(S)| \in \{1, 2, 3, 4, 5\}$; all 35 have $m = 2$ (no Case-B instance with $m = 3$ and $R_Z(S) \neq \emptyset$ was produced by this generator). Representative rows of the output:

p	directions	S	$ S $	$P_\theta(S)$	d_i	$ R_Z(S) $	$\dim U_S$	$ S + 1 - R_Z(S) $	$H_2(S)$	d
3	$(1, 2), (1, 0)$	4×4	16	35	$(1, 1)$	4	13	13	17	$(-1, -1)$
3	$(1, 2), (1, 0)$	hexagon	17	37	$(1, 1)$	5	13	13	18	$(-1, -1)$
3	$(1, 2), (1, -2)$	hexagon	17	45	$(1, 1)$	3	15	15	18	$(-1, -1)$
5	$(1, -3), (1, 0)$	4×4	16	37	$(1, 1)$	2	15	15	17	$(-1, 1)$
2	$(2, -1), (1, 2)$	hexagon	17	41	$(1, 1)$	2	16	16	18	$(-2, 0)$
5	$(2, 1), (1, 2)$	4×4	16	54	$(1, 1)$	1	16	16	17	$(-2, -2)$

On all 35 pairs with $R_Z(S) \neq \emptyset$ the value of $\dim U_S$ was *exactly* $|S| + 1 - |R_Z(S)|$ (over $\mathbb{F}_q$), so the space of affine relations consists of exactly the multiples of A and Lemma 2.5 is tight; and $H_2(S)$ was exactly $\dim U_S + |R_Z(S)| = |S| + 1$, so the quadratic characters make up the deficit exactly, as in §7. On the 116 two-component configurations of Experiment 2 the criterion of Observation C.1 below agreed with the computed case in all 116 instances, and the Case-B direction pairs had $|\det(v_1, v_2)| \in \{2, 3, 4, 5, 6, 7, 10\}$.

Observation C.1 (A criterion for Case B when $m = 2$). Suppose $m = 2$ and both components satisfy $L_i = R_i$; write $\delta_i = F_i - L_i$ for the deviation field, supported in strip i . Replace H_i by a sufficiently large multiple, which by Remark 1.4 does not change the Case A/Case B decision, and expand the four-point template $\{z, z + H_1, z + H_2, z + H_1 + H_2\}$ at an anchor z : δ_1 is constant along the H_1 -pairs and δ_2 along the H_2 -pairs, so $De_\theta(z) = 0$ if and only if at least one of the two

pairs of constants agrees. Hence *Case B holds if and only if* $\text{supp } \delta_1 \cap \text{supp } \delta_2 = \emptyset$, that is, no point is deviated by both components at once. In particular, when $k_i = 1$, the tails are constant and equal, and $|\det(v_1, v_2)| = 1$, one is always in Case A: the map $(\pi_1, \pi_2): \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ is then surjective and the two deviation sets must meet. Case B occurs only for $|\det| \geq 2$. The sample of Experiment 2 was generated at random and then sorted by the numerical test; the criterion explains why Case B appeared only on direction pairs with $|\det| \geq 2$.

C.3. A counterexample to the generalisation of Lemma 3.1. (`lemma31.py`.) Take $p = 2$, $m = 3$, $v_1 = (1, 0)$, $v_2 = (0, -1)$, $v_3 = (1, -1)$. Tails: $L_1 = 0$ and $R_1 = \mathbb{K}[x \text{ even}]$; $L_2 = L_3 = 0$ and $R_2 = R_3 = \mathbb{K}[3 \mid x]$; the transition strip of each component is constant 1, of width 1. Here $\pi_2(v_1) = \pi_3(v_1) = 1 > 0$, so $B_1^+ = R_2 + R_3 \equiv 0$ and $B_1^- = L_2 + L_3 = 0$, and the program computes $\Lambda_1 = \{\pm 1\}$ (with $\kappa_1 = 12$ and frequencies $\{0, 6\}$), so that $A_1(T^{v_1}) = T^{2v_1} - 1$.

- For the pairs of realised sectors adjacent along the rays $\pm v_1$, namely (R_1, R_2, R_3) with (L_1, R_2, R_3) and (R_1, L_2, L_3) with (L_1, L_2, L_3) : the difference after applying $A_1(T^{v_1})$ is 0, up to a floating-point error of $1.3 \cdot 10^{-16}$, for both colours.
- For the mixed choice (R_1, R_2, L_3) against (L_1, R_2, L_3) — here $B = R_2 + L_3 = \mathbb{K}[3 \mid x]$, and the sign vector $(+, +, -)$ corresponds to the cone $\{y > 0, x > 0, x + y < 0\} = \emptyset$, which is not realised: the difference after applying $A_1(T^{v_1})$ has maximum absolute value 2.

By hand: along the x -direction, $\Delta_1 = \mathbb{K}[R_1 + B = 1] - \mathbb{K}[L_1 + B = 1]$ takes the values $(-1, 0, 1, 0, 1, 0)$ for $x \bmod 6 = 0, \dots, 5$, whereas its translate by $x \mapsto x + 2$ is $(1, 0, 1, 0, -1, 0)$. The two differ, so Δ_1 is not killed by $T^{2v_1} - 1$: its spectrum contains cube roots of unity, and those are not in Λ_1 . This is why §3 can only assert that Ae is constant across the *realised* sectors, where $B = B_i^\sigma$ — and that is exactly, and only, what Lemma 3.2 needs.

APPENDIX D. THE $\mathbb{F}_p$ AND CONVEX FORM OF SZABADOS' THEOREM

This appendix proves in full the $\mathbb{F}_p$ and convex version of a theorem of Szabados. Szabados [17, Theorem 1] proved that a sum of two $\mathbb{Z}$ -valued periodic configurations with different period directions does not have low rectangular complexity; the convex generalisation is [2, Theorem 1.7], and the $\mathbb{F}_p$ and convex version is stated, with a sketch, as [5, Theorem 2.8]. Theorem D.1 below is the latter, proved here in full. The proof follows Szabados [17] — pigeonhole, balanced sets, and the one-dimensional Morse–Hedlund theorem — with two changes. The coefficient ring is $\mathbb{F}_p$: every argument below uses only equality of values and the fact that a certain difference is periodic in a given direction, and none of it touches the arithmetic of the ring. And the window is an arbitrary lattice-convex set instead of a rectangle: in Lemma D.5 one deletes, at each step, whichever of the two extreme rows is the shorter. In addition, Lemma 3 of [17] rests on Lemma 39 of Kari and Szabados [11]; we replace it by two self-contained arguments, Lemma D.3, which uses the explicit shape of the annihilator, and step (6) of Lemma D.7, which uses the two-sided recursion supplied by the generating property. Apart from the one-dimensional Morse–Hedlund theorem, nothing external is quoted here.

D.0. The statements.

Theorem D.1. *Let p be a prime, let $\theta_1, \theta_2: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ have periods $h_1, h_2 \in \mathbb{Z}^2 \setminus \{0\}$ respectively, with h_1 and h_2 linearly independent over $\mathbb{R}$, and put $\theta = \theta_1 + \theta_2$. If $P_\theta(S) \leq |S|$ for some non-empty finite lattice-convex S , then θ is periodic.*

Corollary D.2. *Let $\eta: \mathbb{Z}^2 \rightarrow \mathbb{F}_p$ and let $\eta = \eta_1 + \eta_2$ be an $\mathbb{F}_p$ -minimal periodic decomposition, of order 2 — that is, a decomposition into periodic $\mathbb{F}_p$ -configurations with as few summands as possible, here two. Then η does not have low convex complexity.*

Proof. Order 2 forces η to be non-periodic, a periodic configuration having a minimal decomposition of length 1, and it forces the two periods to point in different directions, components with parallel periods being mergeable into one, against minimality. If η had low convex complexity, Theorem D.1 would make it periodic, a contradiction. $\square$

D.1. Notation. Let $\mathbf{u}$ be the primitive vector in the direction of h_1 , so that $h_1 = c_1 \mathbf{u}$ with $c_1 \geq 1$, and let $\pi := \det(\mathbf{u}, \cdot) : \mathbb{Z}^2 \rightarrow \mathbb{Z}$, which is surjective. The *rows* are the fibres $L_t := \pi^{-1}(t)$; neighbouring lattice points of a row differ by $\mathbf{u}$. Put $\Delta := \pi(h_2) \neq 0$ and, replacing h_2 by $-h_2$ if necessary, assume $\Delta > 0$. For integers $a \leq b$ the *strip* $\text{St}[a, b] := \{v : a \leq \pi(v) \leq b\}$, of *width* $b - a$, is the union of the rows $L_a, \dots, L_b$ and is invariant under $\pm \mathbf{u}$. For a non-empty finite $F \subset \mathbb{Z}^2$ put $\text{ht}(F) := \max \pi(F) - \min \pi(F)$; F *fits into* $\text{St}[a, b]$ if some translate of it is contained there, that is if $\text{ht}(F) \leq b - a$. For a non-empty finite lattice-convex F write $E_+(F) := F \cap L_{\max \pi(F)}$ and $E_-(F) := F \cap L_{\min \pi(F)}$, the top and bottom rows; lattice convexity makes every non-empty row of F a run of consecutive lattice points.

For a finite $F \subset \mathbb{Z}^2$ write $\text{Pat}(F, \theta) := \{\theta|_{v+F} : v \in \mathbb{Z}^2\}$, the patterns being regarded as functions $w \mapsto \theta(v + w)$ on F , so that $P_\theta(F) = |\text{Pat}(F, \theta)|$ in the notation of §0.2, with $P_\theta(\emptyset) = 1$; X_θ is the orbit closure, so that every finite-window pattern of an $x \in X_\theta$ is a pattern of θ . “ $x = \theta$ on U ” means $x(v) = \theta(v)$ for all $v \in U$. Inside this appendix the letter D denotes auxiliary finite sets and never the difference operator of §0.3, which does not occur here.

D.2. Two algebraic facts. For $s \in \mathbb{Z}$, the h_2 -periodicity of θ_2 gives

$$\theta(v + sh_2) - \theta(v) = \theta_1(v + sh_2) - \theta_1(v), \quad \text{which is } h_1\text{-periodic in } v. \quad (\text{D.1})$$

Consequently: if $T^{sh_2}\theta$, that is $v \mapsto \theta(v + sh_2)$, agrees with θ on a set F , then it agrees with θ on $F + \mathbb{Z}h_1$.

Lemma D.3 (Periodicity on a strip implies periodicity). *Suppose $b - a \geq \Delta$ and $q \geq 1$, and that $\theta(v + qu) = \theta(v)$ for all $v \in \text{St}[a, b]$. Then θ has period qh_1 .*

Proof. The strip is invariant under $+\mathbf{u}$, so $\theta(v + qh_1) = \theta(v)$ for $v \in \text{St}[a, b]$, applying the hypothesis c_1 times. Put $\psi(v) := \theta(v + qh_1) - \theta(v)$; since θ_1 has period h_1 we have $\psi(v) = \theta_2(v + qh_1) - \theta_2(v)$, so ψ has period h_2 , and $\psi = 0$ on $\text{St}[a, b]$. Write $h_2 = gh'_2$ with h'_2 primitive and take any lattice line $L = \{v_0 + sh'_2 : s \in \mathbb{Z}\}$ in the direction h'_2 . Then $\psi|_L$ is periodic in s with period g , while $\pi(v_0 + sh'_2) = \pi(v_0) + s\pi(h'_2)$ runs through an arithmetic progression of common difference $\pi(h'_2) = \Delta/g \geq 1$; a closed interval $[a, b]$ of length $b - a \geq \Delta = g \cdot (\Delta/g)$ contains at least g of its terms, so L meets $\text{St}[a, b]$ in at least g consecutive of its points. A function of period g vanishing at g consecutive points vanishes identically, so $\psi|_L \equiv 0$. As L was arbitrary, $\psi \equiv 0$. $\square$

D.3. Balanced sets.

Definition D.4. Let B be a non-empty finite lattice-convex set, let $\sigma \in \{+, -\}$ and put $E := E_\sigma(B)$. Call B σ -balanced if

- (i) $P_\theta(B) \leq |B|$;
- (ii) $P_\theta(B) < P_\theta(B \setminus E) + |E|$;
- (iii) every non-empty row of B contains at least $|E| - 1$ points.

Lemma D.5 (Existence). *Let S be a non-empty finite lattice-convex set with $P_\theta(S) \leq |S|$. Then there are a sign σ and a σ -balanced set $B \subseteq S$.*

Proof. Construct $D_0 := S \supsetneq D_1 \supsetneq \cdots \supsetneq D_K = \emptyset$ as follows: if $D_i \neq \emptyset$, choose σ_i with $|E_{\sigma_i}(D_i)| \leq |E_{-\sigma_i}(D_i)|$ — either sign will do when D_i has a single row, where $E_+ = E_- = D_i$ — and put $D_{i+1} := D_i \setminus E_{\sigma_i}(D_i)$. Each D_i is again lattice-convex: after an extreme row has been deleted the remaining points lie in an open half-plane, so the lattice points of their convex hull still belong to D_i and do not lie on the deleted row. Put $f(i) := P_\theta(D_i) - |D_i|$; then $f(0) \leq 0$ and $f(K) = 1$, so there is a smallest i with $f(i+1) > 0$, and for it $f(i) \leq 0 < f(i+1)$. Let $B := D_i$, $\sigma := \sigma_i$ and $E := E_\sigma(B)$. Condition (i) is $f(i) \leq 0$. For (ii), $P_\theta(B) \leq |B| = |B \setminus E| + |E| < P_\theta(B \setminus E) + |E|$.

For (iii), write $n = |E|$ and $E' = E_{-\sigma}(B)$, so $|E'| \geq n$. For a real $t \in [\min \pi(B), \max \pi(B)]$ let $\lambda(t)$ be the length of the intersection of the convex polygon $\text{Conv}(B)$ with the line $\{\pi = t\}$; λ is concave, by the concavity of the length of parallel chords of a convex set. On the two extreme rows that intersection is exactly $\text{Conv}(E)$, respectively $\text{Conv}(E')$, a supporting line meeting the polygon in the convex hull of the vertices it carries, of lengths $(n-1)|u|$ and $(|E'| - 1)|u| \geq (n-1)|u|$. Hence $\lambda(t) \geq (n-1)|u|$ for every t . A closed segment of length at least $(n-1)|u|$ on a lattice line whose consecutive points are $|u|$ apart contains at least $n-1$ lattice points, and these belong to B by lattice convexity. $\square$

D.4. Ambiguity strips and the propagation lemma.

Definition D.6. Say that θ has the $+$ -ambiguity strip $\text{St}[a, b]$, where $a \leq b$, if there is an $x \in X_\theta$ with $x = \theta$ on $\text{St}[a, b-1]$ — an empty condition when $a = b$ — and $x \neq \theta$ at some point of L_b . The $-$ -ambiguity strips are defined symmetrically, replacing u by $-u$.

Lemma D.7 (Propagation). *Let B be $+$ -balanced and put $n := |E_+(B)|$. If θ has some $+$ -ambiguity strip $\text{St}[a, b]$ into which B fits, that is with $b - a \geq \text{ht}(B)$, then θ is periodic.*

Proof. (1) *Placing B .* Translate B so that $\max \pi(B) = b$; then $B \subseteq \text{St}[a, b]$, and $E := E_+(B) = B \cap L_b = \{e_1, \dots, e_n\}$ with $e_{l+1} = e_l + u$, while $B_0 := B \setminus E \subseteq \text{St}[a, b-1]$, possibly empty. Put $t_- := \min \pi(B)$. Take a witness $x \in X_\theta$ of the ambiguity: $x = \theta$ on $\text{St}[a, b-1]$ and $x \neq \theta$ somewhere on L_b . Note that $B_0 + iu \subseteq \text{St}[a, b-1]$ for every $i \in \mathbb{Z}$, so $x = \theta$ on each $B_0 + iu$.

(2) *Two determination rules.* For $0 \leq i \leq n$ put $D_i := B_0 \cup \{e_1, \dots, e_i\}$ and $D'_i := B_0 \cup \{e_{n-i+1}, \dots, e_n\}$. By (ii) of Definition D.4, $P_\theta(D_n) - |D_n| < P_\theta(D_0) - |D_0|$, so there is a $k \in \{0, \dots, n-1\}$ with $P_\theta(D_{k+1}) - |D_{k+1}| < P_\theta(D_k) - |D_k|$, that is $P_\theta(D_{k+1}) < P_\theta(D_k) + 1$; the restriction map $\text{Pat}(D_{k+1}, \theta) \rightarrow \text{Pat}(D_k, \theta)$ being onto, this forces $P_\theta(D_{k+1}) = P_\theta(D_k)$ and makes that map a bijection. Hence:

(R₊) for every $y \in X_\theta$ and $v \in \mathbb{Z}^2$, the value $y(e_{k+1} + v)$ is determined by $y|_{D_k+v}$.

In the same way there is a $k' \in \{0, \dots, n-1\}$ with

(R₋) for every $y \in X_\theta$ and $v \in \mathbb{Z}^2$, the value $y(e_{n-k'} + v)$ is determined by $y|_{D'_{k'}+v}$.

(3) *Claim:* $x|_{D_k+iu} \neq \theta|_{D_k+iu}$ for every $i \in \mathbb{Z}$. Suppose $x = \theta$ on $D_k + i_0u$. If $k = 0$, then $D_0 = B_0$, and (1) together with (R₊) gives $x(e_1 + iu) = \theta(e_1 + iu)$ for every i , that is $x = \theta$ on L_b , against the choice of the witness. So let $k \geq 1$. By (R₊), $x(e_{k+1} + i_0u) = \theta(e_{k+1} + i_0u)$; hence $x = \theta$ on $D_k + (i_0 + 1)u$, whose points are those of $B_0 + (i_0 + 1)u$, by (1), together with the points $e_l + (i_0 + 1)u = e_{l+1} + i_0u$ for $1 \leq l \leq k$, which lie in $D_k + i_0u$ when $l \leq k-1$ and are the point just treated when $l = k$. By induction $x = \theta$ on $D_k + iu$ for every $i \geq i_0$, hence on the half-line $\{e_1 + su : s \geq i_0\} \subset L_b$. Now go left. If $k' = 0$, then (1) and (R₋) give $x = \theta$ on $\{e_n + iu : i \in \mathbb{Z}\} = L_b$, a contradiction. If $k' \geq 1$, choose i_1 with $n - k' + i_1 \geq i_0$; the points $e_1 + (n - k' + i_1)u, \dots, e_1 + (n - 1 + i_1)u$ of the top row of $D'_{k'} + i_1u$ then lie on that half-line, so $x = \theta$ on $D'_{k'} + iu$ for every $i \geq i_1$. By (R₋), $x(e_{n-k'} + i_1u) = \theta(e_{n-k'} + i_1u)$, whence $x = \theta$ on $D'_{k'} + (i_1 - 1)u$, whose top-row points $e_l + (i_1 - 1)u = e_{l-1} + i_1u$, $n - k' + 1 \leq l \leq n$, either

lie in $D'_{k'} + i_1\mathbf{u}$ or are the point just treated. Descending induction gives $x = \theta$ on $D'_{k'} + i\mathbf{u}$ for every $i \in \mathbb{Z}$, and the top rows of these sets cover all of L_b , against the choice of the witness. The claim follows.

(4) *Counting.* By (3) and (1), for each i the patterns $x|_{B+i\mathbf{u}}$ and $\theta|_{B+i\mathbf{u}}$ are two B -patterns of θ which agree on $B_0 + i\mathbf{u}$ and differ somewhere. So every pattern γ in $\Gamma := \{\theta|_{B_0+i\mathbf{u}} : i \in \mathbb{Z}\} \subseteq \text{Pat}(B_0, \theta)$ has at least two extensions to a B -pattern, $N(\gamma) \geq 2$. But

$$\sum_{\gamma \in \text{Pat}(B_0, \theta)} (N(\gamma) - 1) = P_\theta(B) - P_\theta(B_0) < |E| = n$$

by (ii), so $|\Gamma| \leq n - 1$; in particular $n \geq 2$, since otherwise Γ would be empty, which it is not.

(5) *Periodicity of the rows of B_0 .* Assume $B_0 \neq \emptyset$; when $B_0 = \emptyset$ this step is vacuous and one takes $Q_0 := 1$. For each row L_t of B_0 , $t_- \leq t \leq b - 1$, condition (iii) makes $B \cap L_t$ a run of $m_t \geq n - 1$ consecutive points $w_t, w_t + \mathbf{u}, \dots, w_t + (m_t - 1)\mathbf{u}$. Put $\xi_t(s) := \theta(w_t + s\mathbf{u})$ for $s \in \mathbb{Z}$. As i varies, $\theta|_{B_0+i\mathbf{u}}$ takes at most $n - 1$ values, so its restriction to the first $n - 1$ points of row t — that is, the word of length $n - 1$ read in ξ_t from position i — takes at most $n - 1$ values, and ξ_t has at most $n - 1$ subwords of length $n - 1$. By the one-dimensional Morse–Hedlund theorem [13] (a bi-infinite sequence with at most m subwords of some length m is periodic; see also [2, Theorem 1.1]), ξ_t is periodic. Let Q_0 be a common multiple of the periods of these rows; then $\theta(v + Q_0\mathbf{u}) = \theta(v)$ for all $v \in \text{St}[t_-, b - 1]$.

(6) *Extending upwards one row at a time.* Suppose $R \geq b$ and θ has period $Q\mathbf{u}$ on $\text{St}[t_-, R - 1]$, which for $R = b$ is step (5) with $Q = Q_0$. Choose w with $\pi(w) = R - b$, so that the top row of $B + w$ lies on L_R and $B_0 + w + i\mathbf{u} \subseteq \text{St}[t_-, R - 1]$; then the pattern $\theta|_{B_0+w+i\mathbf{u}}$ depends only on $i \bmod Q$. Put $y_s := \theta(e_1 + w + s\mathbf{u})$ and apply (R_+) and (R_-) to $y := \theta$ and $v = w + i\mathbf{u}$:

$$y_{i+k} \text{ is determined by } (i \bmod Q; y_i, \dots, y_{i+k-1}),$$

$$y_{i-1} \text{ is determined by } (i \bmod Q; y_i, \dots, y_{i+k'-1}),$$

the second being (R_-) at $v = w + (i - n + k')\mathbf{u}$. Put $K := \max(k, k', 1)$ and let $\sigma_i := (i \bmod Q; y_i, \dots, y_{i+K-1}) \in \mathbb{Z}/Q \times \mathbb{F}_p^K$. Then σ_{i+1} is determined by σ_i , and so is σ_{i-1} . Let Φ be the forward map, defined on the set Σ of states that actually occur; backward determinacy makes Φ injective on Σ , since $\Phi(\sigma) = \Phi(\sigma')$ forces $\sigma = \sigma'$, and Σ is finite, so Φ is a bijection of Σ and the bi-infinite orbit (σ_i) is periodic. Its period Q' is a multiple of Q , by the phase component, and $y_{s+Q'} = y_s$. Hence θ has period $Q'\mathbf{u}$ on $\text{St}[t_-, R]$. Iterating from $R = b$ up to $R = t_- + \Delta$, finitely many steps, gives a Q^* for which θ has period $Q^*\mathbf{u}$ on the strip $\text{St}[t_-, t_- + \Delta]$, of width Δ .

(7) By Lemma D.3, θ has period Q^*h_1 . □

Remark D.8. Step (5) of Lemma D.7 is the argument of [17, Lemma 4] and of [6, Lemma 2.24]; steps (6) and (7) replace [17, Lemma 3], which appeals to Lemma 39 of [11], and use nothing beyond the generating property and the explicit annihilator.

D.5. Proof of Theorem D.1. Suppose θ is not periodic. Replacing h_2 by ch_2 for an integer $c \geq 1$, which changes neither π nor $\mathbf{u}$, we may assume $\Delta \geq \text{ht}(S) + 1$. Put

$$D := \{\alpha h_1 + \beta h_2 : \alpha, \beta \in [0, 1)\} \cap \mathbb{Z}^2, \quad D_j := D + jh_2, \quad S_j := \text{St}[j\Delta, (j+1)\Delta - 1].$$

Every $v \in \mathbb{Z}^2$ is uniquely $v = \alpha h_1 + \beta h_2$ with $\alpha, \beta \in \mathbb{R}$, and $\pi(v) = \beta\Delta$, so $v \in S_j$ if and only if $\beta \in [j, j+1)$; in that case $v - \lfloor \alpha \rfloor h_1 = \{\alpha\}h_1 + \beta h_2$ is a lattice point of D_j . Hence $S_j = D_j + \mathbb{Z}h_1$, and distinct values of $\lfloor \alpha \rfloor$ give distinct points, so the union is disjoint. Put $N := P_\theta(D) < \infty$.

(a) *A balanced set, and unambiguity.* Take a σ -balanced $B \subseteq S$ by Lemma D.5. We may assume $\sigma = +$: otherwise replace u by $-u$, π by $-\pi$ and h_2 by $-h_2$, which keeps Δ positive, all the definitions being symmetric. Since θ is not periodic, the contrapositive of Lemma D.7 says: *for all $a \leq b$ with $b - a \geq \text{ht}(B)$ and every $x \in X_\theta$, if $x = \theta$ on $\text{St}[a, b - 1]$ then $x = \theta$ on $\text{St}[a, b]$.* Iterating row by row,

$$x = \theta \text{ on } \text{St}[a, b], \quad b - a \geq \text{ht}(B) - 1 \implies x = \theta \text{ on } \{\pi \geq a\}. \quad (\text{D.2})$$

(b) *A witness of non-periodicity.* $N!h_2$ is not a period of θ , so there is a v_0 with $\theta(v_0 + N!h_2) \neq \theta(v_0)$. Put $k := \lfloor \pi(v_0)/\Delta \rfloor - N$, so that $\pi(v_0) \geq (k + N)\Delta$.

(c) *Pigeonhole.* Among the $N + 1$ patterns $\theta|_{D_{k+j}}$, $0 \leq j \leq N$, at most N are distinct, so there are $0 \leq j < j' \leq N$ with $\theta(v + (j' - j)h_2) = \theta(v)$ for all $v \in D_{k+j}$. By (D.1) this holds for all $v \in S_{k+j}$, that is $x := T^{(j'-j)h_2}\theta \in X_\theta$ satisfies $x = \theta$ on $S_{k+j} = \text{St}[(k + j)\Delta, (k + j + 1)\Delta - 1]$. That strip has width $\Delta - 1 \geq \text{ht}(S) \geq \text{ht}(B)$, so (D.2) gives $x = \theta$ on $\{\pi \geq (k + j)\Delta\}$, that is

$$\theta(v + (j' - j)h_2) = \theta(v) \quad \text{whenever } \pi(v) \geq (k + j)\Delta.$$

Since $\pi(h_2) = \Delta > 0$, that half-plane is invariant under $+(j' - j)h_2$; applying the identity repeatedly, and using $(j' - j) \mid N!$, we get $\theta(v + N!h_2) = \theta(v)$ for every v with $\pi(v) \geq (k + j)\Delta$. But $\pi(v_0) \geq (k + N)\Delta \geq (k + j)\Delta$, so $\theta(v_0 + N!h_2) = \theta(v_0)$, contradicting (b). $\square$

Remark D.9. The proof of Theorem D.1 works over any finite alphabet for a sum of two periodic configurations, all that is used being (D.1), that the difference $\theta(\cdot + sh_2) - \theta$ is h_1 -periodic. Over $\mathbb{Z}$ it is the convex generalisation of [17, Theorem 1] recorded as [2, Theorem 1.7].

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Theorem and lemma numbers of the cited works are those of the originals.

REFERENCES

- [1] M. Boyle and D. Lind, *Expansive subdynamics*, Trans. Amer. Math. Soc. **349** (1997), no. 1, 55–102.
- [2] C. F. Colle, *On periodic decompositions, one-sided nonexpansive directions and Nivat’s conjecture*, Discrete Contin. Dyn. Syst. **43** (2023), no. 12, 4299–4327.
- [3] C. F. Colle, *On periodic decompositions and nonexpansive lines*, Math. Z. **303** (2023), art. 80.
- [4] C. F. Colle and E. Garibaldi, *An alphabetical approach to Nivat’s conjecture*, Nonlinearity **33** (2020), 3620–3652.
- [5] C. F. Colle and E. Garibaldi, *A modular structure theorem for minimal periodic decompositions and periodicity of configurations with $P_\eta(4, n) \leq 4n$* , preprint, [arXiv:2606.10193v1](https://arxiv.org/abs/2606.10193v1) [math.DS], 8 June 2026.
- [6] V. Cyr and B. Kra, *Nonexpansive $\mathbb{Z}^2$ -subdynamics and Nivat’s conjecture*, Trans. Amer. Math. Soc. **367** (2015), no. 9, 6487–6537.
- [7] V. Cyr and B. Kra, *Complexity of short rectangles and periodicity*, European J. Combin. **52** (2016), 146–173.
- [8] C. Epifanio, M. Koskas and F. Mignosi, *On a conjecture on bidimensional words*, Theoret. Comput. Sci. **299** (2003), 123–150.
- [9] J. Kari and E. Moutot, *Decidability and periodicity of low complexity tilings*, in 37th International Symposium on Theoretical Aspects of Computer Science (STACS 2020), LIPIcs vol. 154, Art. No. 14, 2020.
- [10] J. Kari and E. Moutot, *Decidability and periodicity of low complexity tilings*, Theory Comput. Syst. **67** (2023), 125–148.
- [11] J. Kari and M. Szabados, *An algebraic geometric approach to Nivat’s conjecture*, Inform. and Comput. **271** (2020), 104481.
- [12] A. Khetan, *A counterexample to Nivat’s conjecture for a non-convex window of full affine span*, preprint, [arXiv:2607.09830v2](https://arxiv.org/abs/2607.09830v2), July 2026.
- [13] M. Morse and G. A. Hedlund, *Symbolic dynamics*, Amer. J. Math. **60** (1938), 815–866.
- [14] M. Nivat, *Invited talk*, 24th International Colloquium on Automata, Languages and Programming (ICALP 1997), Bologna, 1997.

- [15] A. Quas and L. Zamboni, *Periodicity and local complexity*, Theoret. Comput. Sci. **319** (2004), 229–240.
- [16] J. W. Sander and R. Tijdeman, *The rectangle complexity of functions on two-dimensional lattices*, Theoret. Comput. Sci. **270** (2002), 857–863.
- [17] M. Szabados, *Nivat’s conjecture holds for sums of two periodic configurations*, in SOFSEM 2018: Theory and Practice of Computer Science, Lecture Notes in Comput. Sci. vol. 10706, Springer, 2018, pp. 539–551.